Folded Histograms of Irrational Rotations: Sampling Certificates and Structural Mismatch to the Parry Measure

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Abstract

We propose a two-stage audit framework for coarse-grained orbit data and work it out completely in one explicit benchmark. The benchmark is a deterministic fold that sends finite binary words from an irrational rotation to the golden-mean shift by Fibonacci weighting and Zeckendorf normalization. The first stage asks whether an empirical folded histogram is already close to its deterministic folded limit law; the second asks whether the same histogram is compatible with a natural ambient equilibrium, here the Parry block law. In the canonical Sturmian slice $\beta = \alpha$, the induced sequence map is topologically conjugate to the classical higher-block presentation of the Sturmian source, so the fold is dynamically lossless on realized orbit data. This makes the benchmark unusually explicit. We obtain closed formulas for $\mathrm{KL}(\nu\|\pi_m)$, exact support-constrained minima for all Renyi orders, and, for all sufficiently large m, the optimal injective relabeling of the realized Sturmian law into the ambient golden-mean language. The injective optimum is shown to calibrate the universal divergence barrier for all lossless coarse-grainings into primitive shifts of finite type, and the Fibonacci weighting is proved to be the unique choice satisfying a natural contiguous numeration axiom. For bounded-type slopes, the folded law stays within a bounded additive distance of this injective optimum uniformly in the Renyi order, while its divergence from the Parry law still grows like $m \log \varphi - \log m$. On the sampling side, star discrepancy yields explicit total-variation and relative-entropy bounds for empirical folded histograms. By topological conjugacy, the entire audit transfers to nonlinear circle diffeomorphisms with irrational rotation number; a numerical example on the standard circle map confirms this. In the numerical examples, once the first stage of the audit certifies sampling accuracy, the remaining discrepancy with the Parry baseline is seen to be structural rather than statistical.

Key words. irrational rotations, circle maps, Sturmian words, Parry measure, Renyi divergence, discrepancy, shifts of finite type

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1 Introduction

This paper is motivated by a basic question about coarse-grained orbit data. After a deterministic symbolic reduction, when should disagreement with a reference law be attributed to finite sampling, and when does it reflect a mismatch between the source and the reference model itself? Our aim is to make this distinction operational through a two-stage audit. The present paper develops that

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audit in one explicit benchmark: a fold that maps binary rotation words to the golden-mean shift by Fibonacci weighting and Zeckendorf normalization.

Viewed on the ambient cube $\Omega_m = \{0,1\}^m$, the fold is genuinely many-to-one. On the canonical Sturmian slice $\beta = \alpha$, however, its behavior is much more rigid. The realized-support bijection proved below upgrades the fold to a topological conjugacy with the classical m-block presentation of the Sturmian subshift. Thus minimality, unique ergodicity, zero entropy, and linear word complexity are preserved on realized orbit data. In particular, the gap to the Parry measure cannot be blamed on information loss along the fold.

This observation turns the benchmark into a clean instance of the audit framework. On the sampling side, star discrepancy yields explicit total-variation and relative-entropy bounds for empirical microstate laws and their folded histograms; in the Sturmian slice the generic $2m$ count improves to $m+1$. On the equilibrium side, the endpoint structure of the Parry cylinders gives exact formulas for $\mathrm{KL}(\nu\|\pi_m)$, support-constrained minima for all Renyi orders, and, once m is large enough (equivalently $m \geq 6$), the optimal injective relabeling of the realized Sturmian law into X_m. For bounded-type slopes, both the injective optimum and the actual folded law have divergence $m \log \varphi - \log m + O_\alpha(1)$, uniformly in the Renyi order.

These results suggest a simple way to read the numerical experiments. One first checks whether the empirical folded histogram is already close to its deterministic limit law. One then compares the same histogram with the Parry block law. In the examples reported here, the first comparison improves rapidly with the sample size N, whereas the second remains large even when the discrepancy certificate is already tiny. The numerical section is organized to show this two-stage audit in a setting where every term can be computed explicitly.

The ingredients are classical in isolation. Sturmian codings, factor complexity, and factor frequencies go back to [**?**, **?**, **?**] and fit naturally into the substitution and symbolic-dynamical framework [**?**, **?**, **?**]. The Parry measure and the thermodynamic formalism of shifts of finite type are standard [**?**, **?**, **?**]. Fibonacci numeration and finite-state normalization are likewise classical [**?**, **?**, **?**], with related Sturmian and Ostrowski numeration viewpoints in [**?**, **?**]. What is specific to the present model is that these ingredients combine cleanly: the fold is dynamically lossless on the realized Sturmian source, while the comparison with the Parry measure is explicit enough to yield exact finite-m optimization statements.

We deliberately work with one explicit prototype rather than a general theory of fold maps. The advantage is that the main finite-m quantities can be computed exactly and compared directly with numerical data, so the benchmark fully exposes the audit framework in a setting without hidden constants. The limitation is also clear: the sharp formulas rely on the endpoint collapse of Parry cylinders in the golden-mean shift, and we do not claim analogous closed forms for arbitrary shifts of finite type or noncanonical folds. The role of the golden-mean benchmark, however, extends beyond a single example: it calibrates the universal injective-placement barrier for lossless coarse-grainings into primitive shifts of finite type. In general, any deterministic map that is injective on the realized source retains the order-uniform divergence rate $D_q^{\mathrm{opt}} = mh - H_q(\mu_m) + O(1)$. The golden-mean benchmark is exceptional because this universal envelope becomes finite-m exact, the optimizer is explicit, and the actual canonical fold remains within a bounded additive distance of that optimizer. Moreover, by topological conjugacy, the entire audit transfers verbatim to nonlinear circle diffeomorphisms whose rotation number is irrational. For discrepancy and empirical divergence bounds we use [**?**, **?**, **?**, **?**, **?**].

Recent data-driven symbolic methodologies near the present work pursue different objectives. Ordinal and permutation-based methods [**?**] typically use symbolic patterns as robust features for time-series characterization, classification, or transition detection. Symbolic-regression methods [**?**] instead aim to infer governing equations directly from trajectories. Our objective is complementary

to both: the symbolic reduction is fixed a priori, and the question is whether the resulting coarse-grained histogram is already statistically certified and whether it is structurally compatible with a chosen ambient equilibrium. On the numeration side, Ostrowski and related continued-fraction systems [**?**, **?**] provide arithmetic codings with rich dynamical content; the present paper uses the golden branch of that numeration-theoretic viewpoint as an exactly analyzable coarse-graining whose information geometry relative to the Parry law can be solved sharply.

The paper is organized as follows. Section 2 introduces the model, the target laws, and the basic discrepancy setup. Sections 3 and 4 establish the finite-sample estimates and the rigidity results on the Sturmian source. Section 5 contains the comparison with the Parry measure, including the injective optimum, the bounded-type asymptotics, and the uniqueness of the Fibonacci weighting. Section 6 presents the numerical illustrations, including a nonlinear circle-map realization of the audit. Two short appendices collect the optional microstate-level lift identities and a local normalization description of the fold.

2 Model and Target Measures

2.1 Rotations, Windows, and the Folded Observable

Let $\alpha \in (0,1)$ be irrational, let $\beta \in (0,1)$, and let $x_0 \in \mathbb{T} := \mathbb{R}/\mathbb{Z}$. Define the Kronecker orbit

$$x_t := x_0 + t\alpha \pmod 1, \qquad t \geq 0,$$

and the binary readout

$$s_t := \mathbf{1}_{[0,\beta)}(x_t) \in \{0,1\}.$$

For a fixed resolution $m \geq 1$, let

$$\Omega_m := \{0,1\}^m, \qquad X_m := \{u = (u_1, \dots, u_m) \in \{0,1\}^m : \ u_i u_{i+1} = 0 \text{ for } 1 \leq i < m\}.$$

Definition 2.1 (Canonical fold)**.** Let $F_1 = F_2 = 1$ and $F_{k+2} = F_{k+1} + F_k$. For $\omega = \omega_1 \cdots \omega_m \in \Omega_m$, define its Fibonacci value by

$$N(\omega) := \sum_{k=1}^{m} \omega_k F_{k+1}.$$

By Zeckendorf's theorem [**?**], there is a unique sequence $Z(N(\omega)) = (c_1, c_2, \dots) \in \{0,1\}^{\mathbb{N}}$ such that

$$N(\omega) = \sum_{k \geq 1} c_k F_{k+1}, \qquad c_k c_{k+1} = 0 \ \text{ for all } k \geq 1.$$

Let

$$\mathrm{pref}_m(c_1, c_2, \dots) := (c_1, \dots, c_m) \in X_m.$$

The canonical fold at resolution m is

$$\mathrm{Fold}_m(\omega) := \mathrm{pref}_m(Z(N(\omega))) \in X_m.$$

Proposition 2.2 (Basic properties of the canonical fold)**.** *For every $m \geq 1$, the map $\mathrm{Fold}_m : \Omega_m \to X_m$ is well defined, surjective, and satisfies $\mathrm{Fold}_m|_{X_m} = \mathrm{id}$. Moreover, Fold_m admits an order-independent local implementation by value-preserving Zeckendorf rewrites.*

Proof. Well-definedness is immediate from Definition 2.1. If $\omega \in X_m$, then ω is already a legal Zeckendorf prefix, so the uniqueness of the Zeckendorf expansion gives $\mathrm{Fold}_m(\omega) = \omega$. Hence Fold_m fixes X_m pointwise, and surjectivity follows by taking $\omega = x \in X_m$ for any $x \in X_m$. The order-independent local implementation is recorded in Appendix B. □

Proposition 2.3 (Residue characterization of the canonical fold)**.** *For every $m \geq 1$ and $\omega \in \Omega_m$, write*

$$N(\omega) = qF_{m+2} + r, \qquad q \in \{0, 1\}, \qquad 0 \leq r < F_{m+2}.$$

Then

$$\mathrm{Fold}_m(\omega) = \mathrm{pref}_m(Z(r)).$$

Equivalently, for $x \in X_m$,

$$\mathrm{Fold}_m(\omega) = x \iff N(\omega) \equiv N(x) \pmod{F_{m+2}}.$$

Proof. Since

$$N(\omega) \leq \sum_{k=1}^{m} F_{k+1} = F_{m+3} - 2 < F_{m+2} + F_{m+2},$$

the quotient q is either 0 or 1. If $q = 0$, then $r = N(\omega)$ and the claim is immediate.

Assume $q = 1$. Then

$$0 \leq r = N(\omega) - F_{m+2} \leq F_{m+1} - 2 < F_{m+1},$$

so the Zeckendorf expansion $Z(r)$ has no nonzero digit in position m or beyond. Hence appending a 1 in position $m + 1$ produces another legal Zeckendorf digit string, namely the expansion of $N(\omega) = F_{m+2} + r$. Its first m digits are therefore exactly those of $Z(r)$, which proves the first statement.

If $x \in X_m$, then

$$N(x) \leq \sum_{j=0}^{\lfloor (m-1)/2 \rfloor} F_{m-2j+1} < F_{m+2},$$

so $x = \mathrm{pref}_m(Z(N(x)))$. Applying the first statement with $r \equiv N(\omega) \pmod{F_{m+2}}$ yields

$$\mathrm{Fold}_m(\omega) = x \iff \mathrm{pref}_m(Z(r)) = \mathrm{pref}_m(Z(N(x))) \iff r = N(x),$$

which is equivalent to the congruence condition. □

Definition 2.4 (Microstates and folded histograms)**.** For $t \geq 0$, the length-m microstate is

$$\omega_m(t) := s_t s_{t+1} \cdots s_{t+m-1} \in \Omega_m.$$

The folded output is

$$y_m(t) := \mathrm{Fold}_m(\omega_m(t)) \in X_m.$$

For a sample length N, define the empirical microstate law and folded histogram by

$$\hat{\mu}_{m,N}(u) := \frac{1}{N} \sum_{t=0}^{N-1} \mathbf{1}_{\{\omega_m(t)=u\}}, \qquad u \in \Omega_m,$$

$$\hat{\pi}_{m,N}(v) := \frac{1}{N} \sum_{t=0}^{N-1} \mathbf{1}_{\{y_m(t)=v\}}, \qquad v \in X_m.$$

Remark 2.5 (Scope). We fix the canonical fold throughout the paper. Sections 3 and 4 use only that Fold_m is deterministic. The specific Zeckendorf structure enters in the cross-resolution boundary law of Theorem 3.4, in the numerical examples, and in the appendix implementation. The sharp $m+1$ complexity statements in Section 5 use the Sturmian slice $\beta = \alpha$, while Theorem 5.3 is a statement about the Parry reference family on X_m and is independent of the fold once a law on X_m has been specified. This separation is deliberate: Sections 3 and 4 isolate the finite-sample and coarse-graining interfaces of the model, whereas Section 5 is the dynamical core in which the ambient equilibrium structure of the golden-mean shift is combined with the low-complexity support created by the folded rotation coding. The paper does not compare alternative Zeckendorf-based folds: the residue characterization of Proposition 2.3 and the rigidity theorem on realized Sturmian support are specific to the canonical normalization, and no analogous conjugacy statement is claimed here for non-canonical procedures.

Because irrational rotations are uniquely ergodic, the empirical microstate law converges to a deterministic target law induced by Lebesgue measure.

Definition 2.6 (Target laws)**.** Let μ_m be the length-m microstate distribution induced by Lebesgue measure on $\mathbb{T}$:

$$\mu_m(u) := \mathrm{Leb}\{x \in \mathbb{T} : \omega_m(x;0) = u\}, \qquad u \in \Omega_m.$$

Its folded pushforward is

$$\pi_m^{\mathrm{fold}} := (\mathrm{Fold}_m)_* \mu_m \in \Delta(X_m).$$

Remark 2.7 (Spaces, maps, and laws). The basic objects of the paper may be summarized by

$$\begin{array}{ccc} \Omega_m & \xrightarrow{\mathrm{Fold}_m} & X_m \\ \mu_m \downarrow & & \downarrow \pi_m^{\mathrm{fold}} \\ \hat{\mu}_{m,N} & \xrightarrow{(\mathrm{Fold}_m)_*} & \hat{\pi}_{m,N} \end{array} \qquad \begin{array}{ccc} \Omega_{m+1} & \xrightarrow{\mathrm{Fold}_{m+1}} & X_{m+1} \\ \mathrm{pref}_m \downarrow & & \downarrow \rho_{m+1\to m} \\ \Omega_m & \xrightarrow{\mathrm{Fold}_m} & X_m. \end{array}$$

The left diagram separates microstate space Ω_m from folded space X_m. The right diagram records the truncation interface whose failure to commute is measured in Section 3.2 by the projective residual.

2.2 The Parry Measure

As a reference measure on X_m we use the Parry measure of the golden-mean shift. Let

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

The associated two-state Parry chain has transition matrix

$$P = \begin{pmatrix} \varphi^{-1} & \varphi^{-2} \\ 1 & 0 \end{pmatrix},$$

and stationary distribution

$$\pi(1) = \frac{1}{\varphi^2+1}, \qquad \pi(0) = 1 - \pi(1).$$

For $u = u_1 \cdots u_m \in X_m$, define the length-$m$ Parry cylinder law

$$\pi_m(u) := \pi(u_1) \prod_{j=1}^{m-1} P_{u_j u_{j+1}}.$$

We emphasize that π_m is used as a reference measure only. We do not assume that $\pi_m^{\mathrm{fold}} = \pi_m$ for irrational rotations. Since $|X_m| = F_{m+2}$, the ambient shift has topological entropy

$$h_{\mathrm{top}}(X) = \lim_{m\to\infty} \frac{1}{m} \log |X_m| = \log \varphi,$$

and the Parry measure is its unique measure of maximal entropy [**?**, **?**].

Proposition 2.8 (Endpoint collapse of Parry cylinders)**.** *For every legal word $u = u_1 \cdots u_m \in X_m$,*

$$\pi_m(u) = \begin{cases} \pi(0)\varphi^{-(m-1)}, & (u_1, u_m) = (0,0), \\ \pi(0)\varphi^{-m}, & (u_1, u_m) \in \{(0,1),(1,0)\}, \\ \pi(0)\varphi^{-(m+1)}, & (u_1, u_m) = (1,1). \end{cases}$$

Equivalently,

$$\pi_m(u) = \pi(0)\varphi^{-m+\mathbf{1}_{\{u_1=0\}}-\mathbf{1}_{\{u_m=1\}}}.$$

In particular, the Parry cylinder weight depends only on the endpoints of the word.

Proof. Let $r_0 := \varphi$ and $r_1 := 1$. For every legal transition $(i,j) \in \{(0,0),(0,1),(1,0)\}$, one has

$$P_{ij} = \frac{r_j}{\varphi r_i}.$$

Hence for a legal word $u = u_1 \cdots u_m$,

$$\prod_{j=1}^{m-1} P_{u_j u_{j+1}} = \varphi^{-(m-1)} \frac{r_{u_m}}{r_{u_1}}.$$

Multiplying by $\pi(u_1)$ and using $\pi(1) = \pi(0)\varphi^{-2}$ gives the three cases above. □

2.3 Discrepancy of the Rotation Orbit

Our only quantitative input is the one-dimensional star discrepancy of the rotation orbit:

$$D_N^*(\alpha; x_0) := \sup_{0\le a\le 1} \left| \frac{1}{N} \sum_{t=0}^{N-1} \mathbf{1}_{[0,a)}(x_t) - a \right|.$$

This quantity is effective because length-m rotation words are determined by a finite partition of the circle into interval atoms.

Theorem 2.9 (Bounds for Sturmian cylinder lengths)**.** *Let $\alpha = [0; a_1, a_2, \dots]$ and let p_n/q_n be its convergents. If $q_n \le t < q_{n+1}$, then every length-t Sturmian cylinder $C(w)$ satisfies*

$$\frac{1}{q_{n+1}+q_n} \;<\; \mathrm{Leb}(C(w)) \;<\; \frac{1}{q_n} + \frac{1}{q_{n+1}} \;<\; \frac{2}{q_n}.$$

Equivalently,

$$\log\!\left(\frac{q_n}{2}\right) \;<\; -\log \mathrm{Leb}(C(w)) \;<\; \log(q_{n+1}+q_n).$$

If α has bounded partial quotients, then $-\log \mathrm{Leb}(C(w)) = \log t + O(1)$ uniformly in w.

Proof. The endpoints of length-t cylinders are among the orbit points $\{-k\alpha : 0 \le k \le t\}$, so cylinder lengths are the adjacent gaps in this finite rotation set. Standard continued-fraction gap estimates and the three-gap theorem give

$$\min_{1\le k\le t} \|k\alpha\| = \|q_n\alpha\| \in \left(\frac{1}{q_{n+1}+q_n}, \frac{1}{q_{n+1}}\right),$$

while the largest gap is at most $\|q_{n-1}\alpha\| + \|q_n\alpha\| < q_n^{-1} + q_{n+1}^{-1}$; see [**?**, **?**]. Taking negative logarithms yields the second display. □

3 Total Variation Bounds

3.1 Discrepancy Bounds

The first step is to observe that discrepancy controls the empirical mass of each relevant cylinder atom and hence the total variation distance of both the microstate and folded laws.

Theorem 3.1 (Discrepancy control of microstate and folded laws)**.** *In the rotation model above, for every $m, N \ge 1$,*

$$\mathrm{TV}(\hat{\mu}_{m,N}, \mu_m) \le 2m\, D_N^*(\alpha; x_0).$$

Consequently,

$$\mathrm{TV}(\hat{\pi}_{m,N}, \pi_m^{\mathrm{fold}}) \le 2m\, D_N^*(\alpha; x_0).$$

If $\beta = \alpha$, then the sharper bounds

$$\mathrm{TV}(\hat{\mu}_{m,N}, \mu_m) \le (m+1) D_N^*(\alpha; x_0), \qquad \mathrm{TV}(\hat{\pi}_{m,N}, \pi_m^{\mathrm{fold}}) \le (m+1) D_N^*(\alpha; x_0)$$

also hold.

Proof. For fixed m, the length-m rotation words are constant on the interval atoms of the partition of $\mathbb{T}$ cut out by the points

$$0,\ \beta,\ -\alpha,\ \beta-\alpha,\ \ldots,\ -(m-1)\alpha,\ \beta-(m-1)\alpha \pmod 1.$$

There are at most $2m$ such endpoints and therefore at most $2m$ interval atoms. For any interval $I = [a, b) \subset \mathbb{T}$,

$$\left|\frac{1}{N}\sum_{t=0}^{N-1} \mathbf{1}_I(x_t) - \mathrm{Leb}(I)\right| \le \left|\frac{1}{N}\sum_{t=0}^{N-1} \mathbf{1}_{[0,b)}(x_t) - b\right| + \left|\frac{1}{N}\sum_{t=0}^{N-1} \mathbf{1}_{[0,a)}(x_t) - a\right| \le 2D_N^*(\alpha; x_0).$$

Taking half the sum of the atomwise errors gives $\mathrm{TV}(\hat{\mu}_{m,N}, \mu_m) \le 2mD_N^*(\alpha; x_0)$.

If $\beta = \alpha$, the coding is Sturmian and the number of distinct length-m factors is exactly $m+1$ [**?**, **?**]. Summing over the $m+1$ nonempty cylinder atoms yields the sharper constant. The folded estimates follow because deterministic pushforward cannot increase total variation. □

Remark 3.2 (Interpretation). Once $D_N^*(\alpha; x_0)$ is controlled, either arithmetically or numerically, the same estimate immediately bounds the folded histogram. No additional probabilistic approximation is needed.

Corollary 3.3 (Optimal leading constant within the metallic-mean family)**.** *For $A \geq 1$, let*

$$\alpha_A = [0; A, A, \dots], \qquad \lambda_A := \frac{A + \sqrt{A^2+4}}{2}, \qquad \kappa(A) := \frac{A}{\log \lambda_A}.$$

Then there is a constant $C(A) < \infty$ such that for every $N \geq 2$ and every $x_0 \in \mathbb{T}$,

$$D_N^*(\alpha_A; x_0) \leq \frac{\kappa(A)\log N + C(A)}{N}.$$

Consequently, for every fixed resolution $m \geq 1$,

$$\mathrm{TV}(\hat{\pi}_{m,N}, \pi_m^{\mathrm{fold}}) \leq 2m\frac{\kappa(A)\log N + C(A)}{N}.$$

If $\beta = \alpha_A$, this sharpens to

$$\mathrm{TV}(\hat{\pi}_{m,N}, \pi_m^{\mathrm{fold}}) \leq (m+1)\frac{\kappa(A)\log N + C(A)}{N}.$$

Moreover, $\kappa(A)$ is strictly increasing on $[1, \infty)$, so the golden branch $A = 1$ uniquely minimizes the leading $(\log N)/N$ coefficient among metallic means.

Proof. Let p_n/q_n be the convergents of α_A. The standard discrepancy bound [**?**] gives

$$D_N^*(\alpha_A; x_0) \leq \frac{3 + nA}{N} \qquad \text{whenever } q_n \leq N < q_{n+1}.$$

Since $q_{n+1} = Aq_n + q_{n-1}$ and λ_A is the Perron root of $x^2 - Ax - 1 = 0$, one has $q_n \geq \lambda_A^{n-1}$ for $n \geq 1$, hence $n \leq 1 + \log N/\log \lambda_A$. Substituting yields the discrepancy estimate with $C(A) = 3 + A$. The TV bound follows from Theorem 3.1. The monotonicity of $\kappa(A) = A/\log \lambda_A$ on $[1, \infty)$ is elementary, so its minimum is attained uniquely at $A = 1$. □

3.2 Cross-Resolution Residuals

Let $r_{m+1\to m} : \Omega_{m+1} \to \Omega_m$ denote prefix truncation, $r_{m+1\to m}(u_1 \cdots u_{m+1}) = u_1 \cdots u_m$, and let $\rho_{m+1\to m} : X_{m+1} \to X_m$ be the analogous truncation on folded words. This residual is not a second reference metric competing with the Parry comparison. Rather, it isolates a specifically dynamical effect of the canonical fold: finite-resolution aliasing across adjacent window lengths. Define the projective residual

$$E_{m,N} := \mathrm{TV}(\hat{\pi}_{m,N}, (\rho_{m+1\to m})_* \hat{\pi}_{m+1,N}).$$

Theorem 3.4 (Boundary law for the canonical fold)**.** *For the canonical fold of Definition 2.1, let*

$$f_m(\omega) := \mathrm{Fold}_m(r_{m+1\to m}(\omega)), \qquad g_m(\omega) := \rho_{m+1\to m}(\mathrm{Fold}_{m+1}(\omega)),$$

and define the boundary event set

$$\mathcal{B}_{m+1} := \{\omega \in \Omega_{m+1} : f_m(\omega) \neq g_m(\omega)\}.$$

Then

$$E_{m,N} \leq \hat{\mu}_{m+1,N}(\mathcal{B}_{m+1}).$$

Moreover, if $\omega = u_1 \cdots u_{m+1}$ and $v := u_1 \cdots u_m$, then

$$\omega \in \mathcal{B}_{m+1} \iff u_{m+1} = 1 \text{ and } N(v) \geq F_{m+1}.$$

Proof. Both $\hat{\pi}_{m,N}$ and $(\rho_{m+1\to m})_*\hat{\pi}_{m+1,N}$ are pushforwards of $\hat{\mu}_{m+1,N}$, respectively by f_m and g_m. Therefore

$$\mathrm{TV}((f_m)_*\nu, (g_m)_*\nu) \le \nu(f_m \neq g_m)$$

for any probability measure ν, and the first claim follows with $\nu = \hat{\mu}_{m+1,N}$.

For the second claim, write $N := N(v)$. Since

$$0 \le N \le \sum_{k=1}^{m} F_{k+1} = F_{m+3} - 2,$$

the Zeckendorf expansion of N has the form

$$Z(N) = (c_1, \dots, c_{m+1}, 0, 0, \dots).$$

If $u_{m+1} = 0$, then $N(\omega) = N$, so $f_m(\omega) = g_m(\omega)$.

Assume next that $u_{m+1} = 1$, so $N(\omega) = N + F_{m+2}$. If $N < F_{m+1}$, then $c_m = c_{m+1} = 0$, hence

$$Z(N + F_{m+2}) = (c_1, \dots, c_m, 1, 0, 0, \dots)$$

is already a legal Zeckendorf expansion. Therefore

$$g_m(\omega) = \rho_{m+1\to m}(\mathrm{Fold}_{m+1}(\omega)) = (c_1, \dots, c_m) = f_m(\omega),$$

so $\omega \notin \mathcal{B}_{m+1}$.

Conversely, suppose $u_{m+1} = 1$ and $\omega \notin \mathcal{B}_{m+1}$. Write

$$Z(N + F_{m+2}) = (d_1, \dots, d_{m+2}, 0, 0, \dots).$$

Since the first m digits agree, one has $d_j = c_j$ for $1 \le j \le m$. Subtracting the two Zeckendorf expansions gives

$$F_{m+2} = (d_{m+1} - c_{m+1})F_{m+2} + d_{m+2}F_{m+3}.$$

Because $F_{m+3} > F_{m+2}$, necessarily $d_{m+2} = 0$ and $d_{m+1} - c_{m+1} = 1$. Hence $c_{m+1} = 0$ and $d_{m+1} = 1$. Since $(d_1, \dots, d_{m+2})$ is a legal Zeckendorf digit string and $d_m = c_m$, one must also have $c_m = d_m = 0$. Thus $Z(N)$ has no nonzero digit in positions m or $m+1$, which is equivalent to $N < F_{m+1}$.

We have proved

$$\omega \notin \mathcal{B}_{m+1} \iff u_{m+1} = 0 \text{ or } \big(u_{m+1} = 1 \text{ and } N(v) < F_{m+1}\big),$$

which is exactly the stated characterization. □

Proposition 3.5 (Residuals relative to the Parry measure)**.** *Let π_m be the Parry measure on X_m. Since the Parry family is projectively consistent,*

$$\pi_m = (\rho_{m+1\to m})_*\pi_{m+1},$$

one has

$$E_{m,N} \le \mathrm{TV}(\hat{\pi}_{m,N}, \pi_m) + \mathrm{TV}(\hat{\pi}_{m+1,N}, \pi_{m+1}).$$

Proof. Apply the triangle inequality around the pair $(\pi_m, (\rho_{m+1\to m})_*\pi_{m+1})$. □

Corollary 3.6 (Residual comparison bound)**.** *For every $m, N \ge 1$,*

$$E_{m,N} \le (4m+2)D_N^*(\alpha; x_0) + \mathrm{TV}(\pi_m^{\mathrm{fold}}, \pi_m) + \mathrm{TV}(\pi_{m+1}^{\mathrm{fold}}, \pi_{m+1}).$$

If $\beta = \alpha$, one may replace $4m+2$ by $2m+3$.

Proof. Apply Corollary 5.1 at resolutions m and $m+1$ inside Proposition 3.5. The universal coefficients give

$$E_{m,N} \le 2mD_N^*(\alpha; x_0) + 2(m+1)D_N^*(\alpha; x_0) + \mathrm{TV}(\pi_m^{\mathrm{fold}}, \pi_m) + \mathrm{TV}(\pi_{m+1}^{\mathrm{fold}}, \pi_{m+1}),$$

which simplifies to the first claim. In the Sturmian slice, the sharpened coefficients $m+1$ and $m+2$ give $2m+3$. □

4 Relative-Entropy Bounds

4.1 From Total Variation to Relative Entropy

To pass from total variation to relative entropy one needs a lower bound on the smallest atom of the target distribution.

Lemma 4.1 (Relative entropy from total variation under strict positivity)**.** *Let p, q be probability measures on a finite set S, assume $p \ll q$, and suppose that q is strictly positive on S. Let*

$$q_{\min} := \min_{s \in S} q(s) > 0.$$

Then

$$\mathrm{KL}(p\|q) \le \log\left(1 + \frac{\|p-q\|_1^2}{q_{\min}}\right) \le \frac{1}{q_{\min}}\|p-q\|_1^2 = \frac{4}{q_{\min}}\,\mathrm{TV}(p,q)^2.$$

Proof. The standard inequality $\mathrm{KL} \le \log(1+\chi^2)$ gives

$$\mathrm{KL}(p\|q) \le \log\bigl(1+\chi^2(p,q)\bigr), \qquad \chi^2(p,q) := \sum_{s \in S} \frac{(p(s)-q(s))^2}{q(s)}.$$

Since $q(s) \ge q_{\min}$ for every $s \in S$,

$$\chi^2(p,q) \le \frac{1}{q_{\min}} \sum_{s \in S} (p(s)-q(s))^2 \le \frac{1}{q_{\min}}\|p-q\|_1^2.$$

The second inequality uses $\log(1+u) \le u$. See [**?**]. □

4.2 Lower Bounds on the Smallest Rotation Atom

Definition 4.2 (Diophantine lower bound)**.** For a badly approximable irrational α, let $c(\alpha) > 0$ be a constant such that

$$\|k\alpha\| \ge \frac{c(\alpha)}{k} \qquad (k \ge 1),$$

where $\|x\| := \min_{n \in \mathbb{Z}} |x-n|$.

Proposition 4.3 (Lower bound on the smallest microstate atom)**.** *Assume $\beta = \alpha$ and let*

$$q_{\min}(m) := \min\{\mu_m(u) : \mu_m(u) > 0\}.$$

Then

$$q_{\min}(m) \ge \min_{1 \le k \le m} \|k\alpha\| \ge \frac{c(\alpha)}{m}.$$

For the golden slope $\alpha = \varphi^{-1}$ one may take $c(\alpha) = 1/\sqrt{5}$, hence

$$q_{\min}(m) \ge \frac{1}{\sqrt{5}\,m}.$$

Proof. When $\beta = \alpha$, the partition endpoints for length-m microstates reduce to the set $\{-k\alpha : 0 \leq k \leq m\}$. Thus the positive atoms of μ_m are exactly the adjacent gaps determined by these points, and the smallest one is at least $\min_{1\leq k\leq m} \|k\alpha\|$. The badly approximable lower bound then yields $c(\alpha)/m$. For φ^{-1} the optimal Markov constant is $1/\sqrt{5}$. □

4.3 Microstate and Folded Relative-Entropy Bounds

Corollary 4.4 (Microstate relative-entropy bound)**.** *Assume $\beta = \alpha$ and α is badly approximable. Then*

$$\mathrm{KL}(\hat{\mu}_{m,N}\|\mu_m) \leq \frac{4}{q_{\min}(m)}\,\mathrm{TV}(\hat{\mu}_{m,N}, \mu_m)^2 \leq \frac{4(m+1)^2}{q_{\min}(m)} D_N^*(\alpha; x_0)^2.$$

For $\alpha = \varphi^{-1}$ this becomes

$$\mathrm{KL}(\hat{\mu}_{m,N}\|\mu_m) \leq 4\sqrt{5}\, m(m+1)^2 D_N^*(\alpha; x_0)^2.$$

Proof. Let $S_m := \mathrm{supp}(\mu_m)$. In the Sturmian slice $\beta = \alpha$, every observed length-m word belongs to the Sturmian language, hence $\hat{\mu}_{m,N}$ and μ_m may both be viewed as probability measures on S_m. On this set, μ_m is strictly positive and

$$\min_{u\in S_m} \mu_m(u) = q_{\min}(m).$$

The claim therefore follows from Theorem 3.1, Lemma 4.1, and Proposition 4.3. □

Lemma 4.5 (KL monotonicity under pushforward)**.** *Let $F : S \to T$ be a map between finite sets and let p, q be probability measures on S with $p \ll q$. Then*

$$\mathrm{KL}(F_*p\|F_*q) \leq \mathrm{KL}(p\|q).$$

Proof. This is the data-processing inequality for relative entropy under deterministic coarse graining. □

Corollary 4.6 (Folded relative-entropy bound)**.** *Under the assumptions of Corollary 4.4,*

$$\mathrm{KL}(\hat{\pi}_{m,N}\|\pi_m^{\mathrm{fold}}) \leq \mathrm{KL}(\hat{\mu}_{m,N}\|\mu_m) \leq \frac{4(m+1)^2}{q_{\min}(m)} D_N^*(\alpha; x_0)^2.$$

Proof. Apply Lemma 4.5 with $F = \mathrm{Fold}_m$. □

4.4 Sturmian Rigidity on Realized Support

In the canonical Sturmian slice, the fold does not merge the words that are actually realized by the rotation. The reason is arithmetic: along the Sturmian cylinder partition, the Fibonacci values of the factors remain distinct modulo F_{m+2}.

Theorem 4.7 (Sturmian no-collision rigidity)**.** *Assume $\beta = \alpha$, and define*

$$S_m(\alpha) := \mathrm{supp}(\mu_m) \subset \Omega_m, \qquad Y_m := \mathrm{Fold}_m(S_m(\alpha)) \subset X_m.$$

Then the restriction

$$\mathrm{Fold}_m\restriction_{S_m(\alpha)} : S_m(\alpha) \to Y_m$$

is a bijection.

More precisely, let

$$\Xi_m := \{(1-j)\alpha \bmod 1 : \ 0 \le j \le m\} \subset \mathbb{T},$$

let $I_0, \dots, I_m$ be the open atoms of $\mathbb{T} \setminus \Xi_m$ in counterclockwise order, and choose $x_\ell \in I_\ell$. Writing

$$u^{(\ell)} := \omega_m(x_\ell; 0) \in S_m(\alpha), \qquad V_\ell := N(u^{(\ell)}),$$

the cyclic jump multiset

$$\Delta_\ell := V_{\ell+1} - V_\ell \qquad (\ell \pmod{m+1})$$

consists of exactly one positive jump $+F_{m+1}$ and the m negative jumps

$$-1, \ -F_1, \ -F_2, \ \dots, \ -F_{m-1},$$

counted with multiplicity. After a cyclic relabeling that starts immediately to the right of the unique positive jump, the values V_ℓ form a strictly decreasing integer chain. In particular, the residues $V_\ell \bmod F_{m+2}$ are pairwise distinct.

Proof. The proof has two steps: compute the Fibonacci jump produced by crossing each Sturmian boundary atom, and then observe that after the unique positive jump the remaining cycle is a strictly decreasing chain whose total drop is smaller than F_{m+2}, so residue collisions cannot occur. The boundary bookkeeping is the same one that underlies the standard factor partition of Sturmian words; see [**?**, **?**, **?**].

For $0 \le k \le m-1$, the k-th symbol of $\omega_m(x; 0)$ is

$$\mathbf{1}_{[0,\alpha)}(x + k\alpha),$$

so its discontinuities occur at the two points $-k\alpha$ and $\alpha - k\alpha = (1-k)\alpha$ on $\mathbb{T}$. The union of all such discontinuity points is exactly Ξ_m. Hence $\omega_m(x; 0)$ is constant on each atom I_ℓ, and $S_m(\alpha) = \{u^{(\ell)} : 0 \le \ell \le m\}$.

Fix a boundary point $\xi_j := (1-j)\alpha \in \Xi_m$. If $1 \le j \le m-1$, then crossing ξ_j from left to right changes the symbols in positions $j-1$ and j from 01 to 10, while all other positions stay fixed: among the translated points $x + k\alpha$, only the two with $k = j-1$ and $k = j$ meet the coding boundary at the instant $x = \xi_j$. Therefore the Fibonacci value jump is

$$F_{j+1} - F_{j+2} = -F_j.$$

At $\xi_0 = \alpha$, only the first symbol changes, from 1 to 0, so the jump is $-F_1 = -1$. At $\xi_m = (1-m)\alpha$, only the last symbol changes, from 0 to 1, so the jump is $+F_{m+1}$. This proves the stated jump multiset.

Now start with the atom immediately to the right of ξ_m. Along the remaining cycle, every subsequent jump is negative, so the corresponding values form a strictly decreasing integer chain. Since

$$1 + \sum_{j=1}^{m-1} F_j = F_{m+1},$$

the total drop before the cycle closes is exactly $F_{m+1} < F_{m+2}$. Hence no two values in this chain can be congruent modulo F_{m+2}. Proposition 2.3 shows that Fold_m is constant on residue classes modulo F_{m+2}, so its restriction to $S_m(\alpha)$ is injective. Bijectivity onto Y_m is then tautological. $\square$

Remark 4.8 (What the rigidity proof uses). The proof uses three ingredients that are all special to the canonical slice $\beta = \alpha$: the Sturmian partition of the torus into exactly $m+1$ atoms, the explicit Fibonacci weighting in $N(\cdot)$, and the residue characterization of the canonical fold modulo F_{m+2}. The argument is therefore more rigid than a generic coding proof, which is why we do not claim an analogous jumps-and-residues theorem for non-canonical folds.

Corollary 4.9 (Exact relabeling on the realized Sturmian language)**.** *Assume $\beta = \alpha$, and let*

$$B_m := \mathrm{Fold}_m \restriction_{S_m(\alpha)} : S_m(\alpha) \to Y_m.$$

Then B_m is a bijection, and for every $N \geq 1$,

$$\pi_m^{\mathrm{fold}} = (B_m)_* \mu_m, \qquad \hat{\pi}_{m,N} = (B_m)_* \hat{\mu}_{m,N}.$$

Consequently,

$$\mathrm{TV}(\hat{\pi}_{m,N}, \pi_m^{\mathrm{fold}}) = \mathrm{TV}(\hat{\mu}_{m,N}, \mu_m),$$
$$\mathrm{KL}(\hat{\pi}_{m,N} \| \pi_m^{\mathrm{fold}}) = \mathrm{KL}(\hat{\mu}_{m,N} \| \mu_m),$$

and, writing H_q for the Renyi entropy of order q, for every $q \in [0, \infty]$,

$$H_q(\hat{\pi}_{m,N}) = H_q(\hat{\mu}_{m,N}), \qquad H_q(\pi_m^{\mathrm{fold}}) = H_q(\mu_m).$$

Proof. The pushforward identities follow from Theorem 4.7. Since B_m is a bijection between the realized supports, total variation, relative entropy, and all Renyi entropies are invariant under the relabeling B_m. □

4.5 Higher-Block Conjugacy on the Sturmian Source

Theorem 4.10 (Higher-block conjugacy on the Sturmian source)**.** *Assume $\beta = \alpha$, let $X_\alpha \subset \{0,1\}^{\mathbb{Z}}$ be the Sturmian subshift generated by the irrational rotation coding, and fix $m \geq 1$. Define*

$$(\Phi_m(s))_t := \mathrm{Fold}_m(s_t s_{t+1} \cdots s_{t+m-1}), \qquad t \in \mathbb{Z}.$$

Then $\Phi_m : X_\alpha \to \mathcal{Y}_m := \Phi_m(X_\alpha)$ is a topological conjugacy. Equivalently, $\mathcal{Y}_m$ is a letter relabeling of the classical m-block presentation of X_α.

In particular, $\mathcal{Y}_m$ is minimal and uniquely ergodic,

$$h_{\mathrm{top}}(\mathcal{Y}_m) = 0,$$

and, writing $p_Z(n)$ for the word complexity of a subshift Z, one has

$$p_{\mathcal{Y}_m}(n) = p_{X_\alpha}(n+m-1) = n+m \qquad (n \geq 1).$$

Proof. Let $\Psi_m : X_\alpha \to S_m(\alpha)^{\mathbb{Z}}$ be the classical higher-block code

$$(\Psi_m(s))_t := s_t s_{t+1} \cdots s_{t+m-1}.$$

Standard higher-block theory gives that Ψ_m is a topological conjugacy from X_α onto its m-block presentation [**?**]. By Corollary 4.9,

$$B_m := \mathrm{Fold}_m \restriction_{S_m(\alpha)} : S_m(\alpha) \to Y_m$$

is a bijection. Hence the coordinatewise relabeling

$$(\widetilde{B}_m(z))_t := B_m(z_t), \qquad z \in \Psi_m(X_\alpha),$$

is a topological conjugacy from the m-block presentation onto $\mathcal{Y}_m$. Because

$$\Phi_m = \widetilde{B}_m \circ \Psi_m,$$

Φ_m is itself a topological conjugacy.

Minimality and unique ergodicity are preserved under conjugacy, and the irrational rotation coding is classically minimal and uniquely ergodic. The same is therefore true of $\mathcal{Y}_m$. Likewise, topological entropy is preserved, so $h_{\mathrm{top}}(\mathcal{Y}_m) = h_{\mathrm{top}}(X_\alpha) = 0$.

A word of length n in the m-block presentation corresponds to a word of length $n+m-1$ in X_α, and the relabeling $\widetilde{B}_m$ does not change word counts. Therefore

$$p_{\mathcal{Y}_m}(n) = p_{X_\alpha}(n+m-1).$$

Since X_α is Sturmian, $p_{X_\alpha}(k) = k+1$ for every $k \geq 1$, which gives $p_{\mathcal{Y}_m}(n) = n+m$. □

Corollary 4.11 (Pure degeneracy ledger on the Sturmian slice)**.** *Assume $\beta = \alpha$, and let $\mu_m^e := Q_m \pi_m^{\mathrm{fold}}$. Then*

$$\mathrm{KL}(\mu_m \| \mu_m^e) = \mathbb{E}_{\pi_m^{\mathrm{fold}}}\big[\log d_m(Y)\big].$$

Likewise, for every $N \geq 1$, if $\hat{\mu}_{m,N}^e := Q_m \hat{\pi}_{m,N}$, then

$$\mathrm{KL}(\hat{\mu}_{m,N} \| \hat{\mu}_{m,N}^e) = \mathbb{E}_{\hat{\pi}_{m,N}}\big[\log d_m(Y)\big].$$

Proof. Corollary 4.9 gives

$$H(\mu_m) = H(\pi_m^{\mathrm{fold}}), \qquad H(\hat{\mu}_{m,N}) = H(\hat{\pi}_{m,N}).$$

Applying Proposition A.2 to μ_m and $\hat{\mu}_{m,N}$ yields the two identities. □

A complementary microstate lift formalism is collected in Appendix A. It records the fiber-uniform lift family, the exact entropy ledger used in Corollary 4.11, and an optional Pythagorean identity for relative entropy against lifted macro models. In particular, if μ is a microstate law, $\pi := (\mathrm{Fold}_m)_*\mu$, and $Q_m\eta$ denotes the fiber-uniform lift of a macro law η, then Appendix A shows

$$\mathrm{KL}(\mu \| Q_m\eta) = \mathrm{KL}(\pi \| \eta) + \mathrm{KL}(\mu \| Q_m\pi),$$

so the observable macro divergence and the hidden within-fiber residual split exactly. These identities are not needed for the support-barrier results of Section 5, so they are deferred to the appendix in order to keep the main text focused on histogram-level comparison with the Parry measure.

5 Comparison with the Parry Measure

The Parry measure is the unique maximal-entropy equilibrium state of the ambient golden-mean shift, whose topological entropy is $h_{\mathrm{top}}(X) = \log \varphi$. The point of this section is therefore not merely to compare two probability measures on X_m, but to quantify how far folded rotation laws sit from ambient equilibrium when their effective support is much smaller than the full shift language. In computational terms, the Parry law serves here as a reference model against which one can test whether a coarse-grained symbolic representation is merely noisy or structurally incompatible

with the ambient shift geometry. This section is intentionally model-specific: the exact identities below rely on endpoint collapse for the golden-mean Parry cylinders, and no comparable closed formula is claimed here for arbitrary shifts of finite type or non-canonical folds. Within this explicit setting, the bounded-type case sharpens the extensive Parry gap to the order-uniform asymptotic $D_q(\pi_m^{\mathrm{fold}}\|\pi_m) = m\log\varphi - \log m + O_\alpha(1)$. The new point beyond a support budget is that, in the canonical Sturmian slice, this gap persists even after optimizing over all injective recodings of the realized source into the ambient golden-mean language.

Corollary 5.1 (Decomposition relative to the Parry measure)**.** *For every $m, N \geq 1$,*

$$\mathrm{TV}(\hat{\pi}_{m,N}, \pi_m) \leq 2m\, D_N^*(\alpha; x_0) + \mathrm{TV}(\pi_m^{\mathrm{fold}}, \pi_m).$$

If $\beta = \alpha$, one may replace $2m$ by $m+1$.

Proof. Theorem 3.1 gives $\mathrm{TV}(\hat{\pi}_{m,N}, \pi_m^{\mathrm{fold}}) \leq 2mD_N^*(\alpha; x_0)$, and the triangle inequality completes the proof. □

5.1 Exact Parry Identity and Support Barriers

Definition 5.2 (Renyi entropy and divergence)**.** Let E be a finite set. For a probability law ν on E, define

$$H_q(\nu) := \begin{cases} \log|\operatorname{supp}(\nu)|, & q = 0, \\ \dfrac{1}{1-q}\log\Bigl(\sum_{x\in E}\nu(x)^q\Bigr), & q \in (0,\infty)\setminus\{1\}, \\ -\sum_{x\in E}\nu(x)\log\nu(x), & q = 1, \\ -\log\max_{x\in E}\nu(x), & q = \infty. \end{cases}$$

For probability laws ν and ρ on E, define

$$D_q(\nu\|\rho) := \begin{cases} -\log\rho(\operatorname{supp}(\nu)), & q = 0, \\ \dfrac{1}{q-1}\log\Bigl(\sum_{x\in E}\nu(x)^q\rho(x)^{1-q}\Bigr), & q \in (0,\infty)\setminus\{1\}, \\ \mathrm{KL}(\nu\|\rho), & q = 1, \\ \log\max_{x:\,\nu(x)>0}\dfrac{\nu(x)}{\rho(x)}, & q = \infty, \end{cases}$$

with the convention $0\log 0 := 0$, and with $D_q(\nu\|\rho) := \infty$ for $q > 0$ if ν is not absolutely continuous with respect to ρ.

Theorem 5.3 (Exact divergence identity relative to the Parry measure)**.** *Let ν be a probability measure on X_m, and define the endpoint masses*

$$\eta_{ij}(\nu) := \nu\bigl(\{u \in X_m : u_1 = i,\ u_m = j\}\bigr), \qquad i, j \in \{0,1\}.$$

If, with the convention $0\log 0 := 0$,

$$H(\nu) := -\sum_{u\in X_m}\nu(u)\log\nu(u)$$

denotes the Shannon entropy of ν, then

$$\mathrm{KL}(\nu\|\pi_m) = m\log\varphi - H(\nu) - \log\pi(0) + (\eta_{11}(\nu) - \eta_{00}(\nu))\log\varphi.$$

Consequently,

$$m\log\varphi - H(\nu) - \log\pi(0) - \log\varphi \le \mathrm{KL}(\nu\|\pi_m) \le m\log\varphi - H(\nu) - \log\pi(0) + \log\varphi.$$

Proof. By Proposition 2.8,

$$\log\pi_m(u) = \log\pi(0) - m\log\varphi + \big(\mathbf{1}_{\{u_1=0\}} - \mathbf{1}_{\{u_m=1\}}\big)\log\varphi.$$

Therefore

$$\begin{aligned}\mathrm{KL}(\nu\|\pi_m) &= \sum_{u\in X_m}\nu(u)\log\frac{\nu(u)}{\pi_m(u)}\\ &= -H(\nu) - \sum_{u\in X_m}\nu(u)\log\pi_m(u)\\ &= m\log\varphi - H(\nu) - \log\pi(0) + \big(\eta_{11}(\nu) - \eta_{00}(\nu)\big)\log\varphi.\end{aligned}$$

Since $\eta_{11}(\nu) - \eta_{00}(\nu) \in [-1, 1]$, the two-sided bound follows. □

Remark 5.4 (Specificity of the endpoint collapse). Theorem 5.3 depends essentially on Proposition 2.8. For a general shift of finite type, Parry cylinder weights are controlled by boundary states in an edge presentation but need not collapse to a scalar endpoint statistic of the kind available here. Accordingly, we do not claim an equally sharp closed formula outside collapse geometries of this type; the exact identity is part of what makes the golden-mean setting especially rigid.

Corollary 5.5 (No fold-entropy loss on the Sturmian slice)**.** *Assume $\beta = \alpha$. Then*

$$\mathrm{KL}(\pi_m^{\mathrm{fold}}\|\pi_m) = m\log\varphi - H(\mu_m) - \log\pi(0) + \big(\eta_{11}(\pi_m^{\mathrm{fold}}) - \eta_{00}(\pi_m^{\mathrm{fold}})\big)\log\varphi.$$

Likewise, for every $N \ge 1$,

$$\mathrm{KL}(\hat\pi_{m,N}\|\pi_m) = m\log\varphi - H(\hat\mu_{m,N}) - \log\pi(0) + \big(\eta_{11}(\hat\pi_{m,N}) - \eta_{00}(\hat\pi_{m,N})\big)\log\varphi.$$

Proof. Apply Theorem 5.3 with $\nu = \pi_m^{\mathrm{fold}}$ and $\nu = \hat\pi_{m,N}$, and use Corollary 4.9 to replace $H(\pi_m^{\mathrm{fold}})$ by $H(\mu_m)$ and $H(\hat\pi_{m,N})$ by $H(\hat\mu_{m,N})$. □

Proposition 5.6 (Order-uniform Renyi window relative to the Parry measure)**.** *For every probability law ν on X_m and every $q \in [0, \infty]$,*

$$m\log\varphi - H_q(\nu) - \log\pi(0) - \log\varphi \le D_q(\nu\|\pi_m) \le m\log\varphi - H_q(\nu) - \log\pi(0) + \log\varphi.$$

Proof. For $q = 1$, this is Theorem 5.3. For $q = 0$, let $S := \mathrm{supp}(\nu)$. Proposition 2.8 gives

$$\pi(0)\varphi^{-(m+1)} \le \pi_m(u) \le \pi(0)\varphi^{-(m-1)} \qquad (u \in X_m),$$

and therefore

$$|S|\pi(0)\varphi^{-(m+1)} \le \pi_m(S) \le |S|\pi(0)\varphi^{-(m-1)}.$$

Since $D_0(\nu\|\pi_m) = -\log\pi_m(S)$ and $H_0(\nu) = \log|S|$, the displayed bounds follow.

For $q = \infty$, the same pointwise bounds on $\pi_m(u)$ imply

$$\frac{\varphi^{m-1}}{\pi(0)} \max_{u \in X_m} \nu(u) \le \max_{u:\, \nu(u)>0} \frac{\nu(u)}{\pi_m(u)} \le \frac{\varphi^{m+1}}{\pi(0)} \max_{u \in X_m} \nu(u).$$

Taking logarithms and using $H_\infty(\nu) = -\log \max_u \nu(u)$ gives the claim.

Now fix $q \in (0, \infty) \setminus \{1\}$. By Proposition 2.8,

$$\pi_m(u) = \pi(0)\varphi^{-m+\theta(u)}, \qquad \theta(u) := \mathbf{1}_{\{u_1=0\}} - \mathbf{1}_{\{u_m=1\}} \in [-1, 1].$$

Hence

$$\sum_{u \in X_m} \nu(u)^q \pi_m(u)^{1-q} = \pi(0)^{1-q} \varphi^{m(q-1)} \sum_{u \in X_m} \nu(u)^q \varphi^{(1-q)\theta(u)}.$$

Because $\theta(u) \in [-1, 1]$, one has

$$\varphi^{-|1-q|} \sum_{u \in X_m} \nu(u)^q \le \sum_{u \in X_m} \nu(u)^q \varphi^{(1-q)\theta(u)} \le \varphi^{|1-q|} \sum_{u \in X_m} \nu(u)^q.$$

Taking logarithms, dividing by $q - 1$, and using the definition of H_q gives the same two-sided bound with error at most $\log \varphi$. □

Proposition 5.7 (Support conditioning for Renyi divergence)**.** *Let E be finite, let ρ be a probability law on E, and let $S \subset \operatorname{supp}(\rho)$ satisfy $\rho(S) > 0$. Writing $\rho^S := \rho(\cdot \mid S)$, every probability law ν on E with $\operatorname{supp}(\nu) \subset S$ satisfies*

$$D_q(\nu \| \rho) = D_q(\nu \| \rho^S) - \log \rho(S) \qquad (q \in [0, \infty]).$$

Consequently,

$$\inf_{\substack{\nu \in \Delta(E) \\ \operatorname{supp}(\nu) \subset S}} D_q(\nu \| \rho) = -\log \rho(S) \qquad (q \in [0, \infty]).$$

If $q \in (0, \infty]$, the minimizer is unique and equals ρ^S. If $q = 0$, the minimizers are exactly the laws whose support is S.

Proof. For $q = 1$, the identity is the usual conditioning formula for relative entropy. For $q = 0$,

$$D_0(\nu \| \rho) = -\log \rho(\operatorname{supp}(\nu)) = -\log \rho^S(\operatorname{supp}(\nu)) - \log \rho(S) = D_0(\nu \| \rho^S) - \log \rho(S).$$

For $q = \infty$,

$$D_\infty(\nu \| \rho) = \log \max_{x:\, \nu(x)>0} \frac{\nu(x)}{\rho(x)} = \log \max_{x:\, \nu(x)>0} \frac{\nu(x)}{\rho(S)\rho^S(x)} = D_\infty(\nu \| \rho^S) - \log \rho(S).$$

Finally, for $q \in (0, \infty) \setminus \{1\}$,

$$\sum_{x \in E} \nu(x)^q \rho(x)^{1-q} = \sum_{x \in S} \nu(x)^q \big(\rho(S)\rho^S(x)\big)^{1-q} = \rho(S)^{1-q} \sum_{x \in S} \nu(x)^q (\rho^S(x))^{1-q},$$

which gives the claimed identity after taking logarithms and dividing by $q - 1$.

For every $q > 0$, the standard nonnegativity $D_q(\nu \| \rho^S) \ge 0$ shows that $D_q(\nu \| \rho) \ge -\log \rho(S)$. Equality holds exactly when $D_q(\nu \| \rho^S) = 0$, namely when $\nu = \rho^S$. For $q = 0$,

$$D_0(\nu \| \rho^S) = -\log \rho^S(\operatorname{supp}(\nu)) \ge 0,$$

with equality if and only if $\rho^S(\operatorname{supp}(\nu)) = 1$, equivalently $\operatorname{supp}(\nu) = S$. □

Lemma 5.8 (Endpoint class counts)**.** *Extend the Fibonacci sequence by $F_0 := 0$, $F_1 := 1$, and $F_{n+2} = F_{n+1} + F_n$. For $m \geq 2$, let*

$$X_m^{ij} := \{u \in X_m : u_1 = i,\ u_m = j\}, \qquad i, j \in \{0, 1\}.$$

Then

$$|X_m^{00}| = F_m, \qquad |X_m^{01}| = |X_m^{10}| = F_{m-1}, \qquad |X_m^{11}| = F_{m-2}.$$

Proof. The admissible length-m words in X_m are the paths of length $m-1$ in the golden-mean graph with adjacency matrix

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

For the standard Fibonacci extension above,

$$A^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \qquad (n \geq 1).$$

Hence the number of admissible words of length m with endpoints (i, j) is $(A^{m-1})_{ij}$, which gives the displayed formulas. □

Theorem 5.9 (Sharp support barrier against the Parry measure)**.** *Let $m \geq 2$ and $1 \leq k \leq F_{m+2}$. Define*

$$M_m(k) := \max\{\pi_m(S) : S \subset X_m,\ |S| \leq k\}.$$

Then

$$M_m(k) = \begin{cases} k\,\pi(0)\varphi^{-(m-1)}, & 1 \leq k \leq F_m, \\ F_m\pi(0)\varphi^{-(m-1)} + (k - F_m)\pi(0)\varphi^{-m}, & F_m < k \leq F_m + 2F_{m-1}, \\ 1 - (F_{m+2} - k)\pi(0)\varphi^{-(m+1)}, & F_m + 2F_{m-1} < k \leq F_{m+2}. \end{cases}$$

Moreover,

$$\inf_{\substack{\nu \in \Delta(X_m) \\ |\operatorname{supp}(\nu)| \leq k}} D_q(\nu \| \pi_m) = -\log M_m(k), \qquad q \in [0, \infty],$$

and

$$\inf_{\substack{\nu \in \Delta(X_m) \\ |\operatorname{supp}(\nu)| \leq k}} \mathrm{TV}(\nu, \pi_m) = 1 - M_m(k).$$

If $q \in (0, \infty]$, the Renyi infimum is attained uniquely by conditioning π_m on a maximizing set $S \subset X_m$; equivalently, one takes all words from the heaviest endpoint classes until the budget k is exhausted. If $q = 0$, the minimizers are exactly the laws whose support is such a maximizing set. The same conditioned Parry laws attain the TV infimum.

Proof. The argument has three layers. One first orders individual Parry cylinder weights by endpoint class, then maximizes the mass of a k-point set by a greedy filling of these endpoint classes, and finally invokes Proposition 5.7 to identify the optimal Renyi law on a fixed support.

By Proposition 2.8, every word in X_m^{00} has Parry weight $\pi(0)\varphi^{-(m-1)}$, every word in $X_m^{01} \cup X_m^{10}$ has weight $\pi(0)\varphi^{-m}$, and every word in X_m^{11} has weight $\pi(0)\varphi^{-(m+1)}$. These three values are strictly ordered. Together with Lemma 5.8, this implies that the largest possible Parry mass of a k-point subset is obtained greedily by exhausting the endpoint classes in the order

$$X_m^{00}, \qquad X_m^{01} \cup X_m^{10}, \qquad X_m^{11},$$

which gives the formula for $M_m(k)$.

Now let $S \subset X_m$ and let ν be supported on S. By Proposition 5.7,

$$D_q(\nu\|\pi_m) \geq -\log \pi_m(S) \qquad (q \in [0,\infty]),$$

with equality for the conditioned law $\pi_m^S := \pi_m(\cdot \mid S)$, and with uniqueness when $q \in (0,\infty]$. Therefore

$$\inf_{\substack{\nu\in\Delta(X_m)\\ |\operatorname{supp}(\nu)|\leq k}} D_q(\nu\|\pi_m) = -\log \max_{\substack{S\subset X_m\\ |S|\leq k}} \pi_m(S) = -\log M_m(k)$$

for every $q \in [0,\infty]$. The characterization of minimizers follows from the same proposition.

For total variation, if ν is supported on S, then

$$\mathrm{TV}(\nu,\pi_m) \geq \pi_m(S^c) = 1 - \pi_m(S).$$

Equality holds for $\nu = \pi_m^S$, since then

$$\mathrm{TV}(\pi_m^S,\pi_m) = \frac{1}{2}\sum_{u\in X_m} \left|\pi_m^S(u) - \pi_m(u)\right| = 1 - \pi_m(S).$$

Taking the best support set S gives the total-variation formula. □

Corollary 5.10 (Exact complexity-divergence law)**.** *Let $m \geq 2$ and $1 \leq k \leq F_m$. Then*

$$\inf_{\substack{\nu\in\Delta(X_m)\\ |\operatorname{supp}(\nu)|\leq k}} D_q(\nu\|\pi_m) = (m-1)\log\varphi - \log\pi(0) - \log k, \qquad q \in [0,\infty],$$

and

$$\inf_{\substack{\nu\in\Delta(X_m)\\ |\operatorname{supp}(\nu)|\leq k}} \mathrm{TV}(\nu,\pi_m) = 1 - k\pi(0)\varphi^{-(m-1)}.$$

If $q \in (0,\infty]$, the Renyi minimizers are exactly the uniform laws on k-point subsets of X_m^{00}. If $q = 0$, the minimizers are exactly the laws whose support is such a k-point subset. The TV minimizers are the conditioned Parry laws on these same subsets.

Consequently, if $k_m \leq F_m$ and

$$\lim_{m\to\infty} \frac{1}{m}\log k_m = \alpha \in [0,\log\varphi],$$

then

$$\lim_{m\to\infty} \inf_{\substack{\nu\in\Delta(X_m)\\ |\operatorname{supp}(\nu)|\leq k_m}} \frac{1}{m} D_q(\nu\|\pi_m) = \log\varphi - \alpha = h_{\mathrm{top}}(X) - \alpha, \qquad q \in [0,\infty].$$

In particular, every subexponential support budget still incurs the full rate $\log\varphi$, and cancelling the Parry penalty requires support growth of order φ^m.

Proof. The first branch of Theorem 5.9 gives the displayed formulas. The description of the minimizers follows because all words in X_m^{00} carry the same Parry weight $\pi(0)\varphi^{-(m-1)}$, so the conditioned Parry law on any k-point subset of X_m^{00} is uniform. The limit statement is obtained by dividing the Renyi formula by m and letting $m \to \infty$. □

5.2 Optimal Injective Relabeling of the Realized Sturmian Law

Theorem 5.11 (Optimal injective relabeling of the realized Sturmian law)**.** *Assume $\beta = \alpha$ and $m \geq 6$. Let $S_m(\alpha)$ be the realized Sturmian language of length m, let μ_m be its factor law, and define*

$$\mathcal{J}_m := \{\iota : S_m(\alpha) \hookrightarrow X_m : \ \iota \text{ injective}\}, \qquad L_q(m) := \inf_{\iota \in \mathcal{J}_m} D_q(\iota_* \mu_m \| \pi_m).$$

Then for every $q \in [0, \infty]$,

$$L_q(m) = (m-1)\log\varphi - \log\pi(0) - H_q(\mu_m).$$

Every injection whose image is contained in X_m^{00} attains the infimum. Moreover, the canonical folded law satisfies

$$0 \leq D_q(\pi_m^{\text{fold}} \| \pi_m) - L_q(m) \leq 2\log\varphi, \qquad q \in [0, \infty],$$

and for $q = 1$,

$$\mathrm{KL}(\pi_m^{\text{fold}} \| \pi_m) - L_1(m) = \big(1 + \eta_{11}(\pi_m^{\text{fold}}) - \eta_{00}(\pi_m^{\text{fold}})\big)\log\varphi.$$

Proof. Let

$$c_m := \pi(0)\varphi^{-(m-1)}.$$

By Proposition 2.8, every word in X_m^{00} has Parry mass c_m, and no word in X_m has larger Parry mass. By Lemma 5.8,

$$|X_m^{00}| = F_m.$$

Since $|S_m(\alpha)| = m+1$ and $F_m \geq m+1$ for $m \geq 6$, there exist injections $\iota \in \mathcal{J}_m$ with $\iota(S_m(\alpha)) \subset X_m^{00}$.

Fix such an injection and write $S := \iota(S_m(\alpha))$. Then $|S| = m+1$, $\pi_m(S) = (m+1)c_m$, and the conditioned Parry law π_m^S is uniform on S. Because $\iota_*\mu_m$ is a relabeling of μ_m, it has the same Renyi entropies. Therefore

$$D_q(\iota_*\mu_m \| \pi_m^S) = \log|S| - H_q(\mu_m) \qquad (q \in [0, \infty]),$$

and Proposition 5.7 gives

$$D_q(\iota_*\mu_m \| \pi_m) = \log(m+1) - H_q(\mu_m) - \log((m+1)c_m) = -\log c_m - H_q(\mu_m).$$

This proves the upper bound

$$L_q(m) \leq -\log c_m - H_q(\mu_m) = (m-1)\log\varphi - \log\pi(0) - H_q(\mu_m).$$

For the matching lower bound, fix an arbitrary $\iota \in \mathcal{J}_m$, write $p_u := \mu_m(u)$, and set $\rho_u := \pi_m(\iota(u))$. Since $\rho_u \leq c_m$ for every $u \in S_m(\alpha)$, the Shannon case satisfies

$$D_1(\iota_*\mu_m \| \pi_m) = \sum_{u \in S_m(\alpha)} p_u \log\frac{p_u}{\rho_u} \geq \sum_{u \in S_m(\alpha)} p_u \log\frac{p_u}{c_m} = -H_1(\mu_m) - \log c_m.$$

For $q \in (0, \infty) \setminus \{1\}$, monotonicity of $x \mapsto x^{1-q}$ implies

$$\frac{1}{q-1}\log\Big(\sum_{u \in S_m(\alpha)} p_u^q \rho_u^{1-q}\Big) \geq \frac{1}{q-1}\log\Big(c_m^{1-q} \sum_{u \in S_m(\alpha)} p_u^q\Big),$$

because $q > 1$ gives a direct lower bound inside the logarithm while $q < 1$ gives an upper bound inside the logarithm and a negative prefactor. Hence

$$D_q(\iota_*\mu_m \| \pi_m) \geq -\log c_m - H_q(\mu_m) \qquad \big(q \in (0, \infty) \setminus \{1\}\big).$$

For $q = \infty$,

$$D_\infty(\iota_*\mu_m\|\pi_m) = \log \max_{u\in S_m(\alpha)} \frac{p_u}{\rho_u} \geq \log \max_{u\in S_m(\alpha)} \frac{p_u}{c_m} = -\log c_m - H_\infty(\mu_m).$$

Finally, since $|S_m(\alpha)| = m + 1$,

$$\pi_m(\iota(S_m(\alpha))) \leq (m+1)c_m = c_m e^{H_0(\mu_m)},$$

and therefore

$$D_0(\iota_*\mu_m\|\pi_m) = -\log \pi_m(\iota(S_m(\alpha))) \geq -\log c_m - H_0(\mu_m).$$

The lower bound is now proved for every $q \in [0, \infty]$, so the formula for $L_q(m)$ follows. Every injection into X_m^{00} attains equality by the first part of the argument.

The left inequality

$$0 \leq D_q(\pi_m^{\text{fold}}\|\pi_m) - L_q(m)$$

is immediate because $\pi_m^{\text{fold}} = (B_m)_*\mu_m$ with $B_m \in \mathcal{J}_m$ by Corollary 4.9. The upper bound follows from the Renyi invariance

$$H_q(\pi_m^{\text{fold}}) = H_q(\mu_m)$$

and Proposition 5.6, which give

$$D_q(\pi_m^{\text{fold}}\|\pi_m) \leq m \log \varphi - H_q(\mu_m) - \log \pi(0) + \log \varphi = L_q(m) + 2 \log \varphi.$$

For $q = 1$, Corollary 5.5 yields

$$\mathrm{KL}(\pi_m^{\text{fold}}\|\pi_m) = m \log \varphi - H_1(\mu_m) - \log \pi(0) + \big(\eta_{11}(\pi_m^{\text{fold}}) - \eta_{00}(\pi_m^{\text{fold}})\big) \log \varphi,$$

so subtracting the displayed formula for $L_1(m)$ gives the exact identity. □

Remark 5.12 (Injective relabeling versus support budgets). Theorem 5.11 does more than constrain the image support size. It fixes the realized Sturmian weight vector μ_m and allows only injective placements of those $m + 1$ probabilities inside the ambient language X_m. The resulting optimum is therefore a coding invariance statement for the realized source, not just a best $(m + 1)$-state approximation problem over arbitrary laws on X_m. The finite exceptional cases $m < 6$ are irrelevant for the asymptotic corollaries below and are absorbed there into the $O_\alpha(1)$ constants.

5.3 Support Budgets and Bounded-Type Asymptotics

Proposition 5.13 (Bounded-type Sturmian Renyi entropy)**.** *Assume $\beta = \alpha$, and suppose that α has bounded partial quotients, equivalently that α is badly approximable in dimension one [**?**, **?**]. Then there exist constants $c_\alpha, C_\alpha > 0$ such that for every $m \geq 1$ and every $u \in S_m(\alpha)$,*

$$\frac{c_\alpha}{m} \leq \mu_m(u) \leq \frac{C_\alpha}{m}.$$

Consequently,

$$H_q(\mu_m) = \log m + O_\alpha(1) \qquad (q \in [0, \infty]),$$

with the $O_\alpha(1)$ term uniform in q.

Proof. Let p_n/q_n be the convergents of α, and choose n so that $q_n \le m < q_{n+1}$. By Theorem 2.9, every length-m Sturmian cylinder satisfies

$$\frac{1}{q_{n+1}+q_n} < \mu_m(u) < \frac{2}{q_n}.$$

If the partial quotients of α are bounded by A, then $q_{n+1} \le (A+1)q_n$, and since $q_n \le m < q_{n+1}$, one has

$$\frac{1}{(A+2)m} \le \mu_m(u) \le \frac{2(A+1)}{m}.$$

This proves the uniform m^{-1} bounds.

Because $|S_m(\alpha)| = m+1$, one has

$$H_0(\mu_m) = \log(m+1), \qquad H_\infty(\mu_m) = -\log \max_{u \in S_m(\alpha)} \mu_m(u) \ge \log m - \log C_\alpha.$$

Since Renyi entropy on a finite alphabet is nonincreasing in q,

$$\log m - \log C_\alpha \le H_q(\mu_m) \le \log(m+1) \qquad (q \in [0,\infty]).$$

This yields the claimed uniform asymptotic. □

Proposition 5.14 (Explicit bounded-type Renyi window)**.** *Assume $\beta = \alpha$, and suppose that the partial quotients of α are bounded by A. Then for every $m \ge 1$ and every $q \in [0,\infty]$,*

$$\log m - \log(2(A+1)) \le H_q(\mu_m) \le \log(m+1).$$

Consequently,

$$m \log \varphi - \log(m+1) - \log \pi(0) - \log \varphi \le D_q(\pi_m^{\mathrm{fold}} \| \pi_m)$$
$$\le m \log \varphi - \log m + \log(2(A+1)) - \log \pi(0) + \log \varphi.$$

In particular,

$$D_q(\pi_m^{\mathrm{fold}} \| \pi_m) - \big(m \log \varphi - \log m\big)$$

remains bounded by explicit constants depending only on A and $\pi(0)$, uniformly in q.

Proof. The proof of Proposition 5.13 gives

$$\frac{1}{(A+2)m} \le \mu_m(u) \le \frac{2(A+1)}{m} \qquad (u \in S_m(\alpha)).$$

Because $|S_m(\alpha)| = m+1$, one has

$$H_0(\mu_m) = \log(m+1),$$

while the pointwise upper mass bound yields

$$H_\infty(\mu_m) = -\log \max_{u \in S_m(\alpha)} \mu_m(u) \ge \log m - \log(2(A+1)).$$

By monotonicity in q,

$$\log m - \log(2(A+1)) \le H_q(\mu_m) \le \log(m+1) \qquad (q \in [0,\infty]).$$

Corollary 4.9 gives

$$H_q(\pi_m^{\mathrm{fold}}) = H_q(\mu_m) \qquad (q \in [0,\infty]),$$

and Proposition 5.6 then yields

$$D_q(\pi_m^{\mathrm{fold}} \| \pi_m) = m \log \varphi - H_q(\mu_m) - \log \pi(0) + O(1),$$

where the $O(1)$ term is bounded in absolute value by $\log \varphi$, uniformly in q. Combining these estimates gives the displayed window. □

Corollary 5.15 (Injective-encoding invariance of the Parry barrier)**.** *Assume $\beta = \alpha$, and suppose that α has bounded partial quotients. Then*

$$\inf_{\iota \in \mathcal{J}_m} D_q(\iota_* \mu_m \| \pi_m) = m \log \varphi - \log m + O_\alpha(1), \qquad q \in [0,\infty],$$

where the $O_\alpha(1)$ term may be chosen uniformly in q. Equivalently, every injective encoding of the realized Sturmian law into X_m has the same linear Parry penalty $\log \varphi$ and the same logarithmic second-order term $-\log m$ up to bounded constants. In particular,

$$D_q(\pi_m^{\mathrm{fold}} \| \pi_m) = \inf_{\iota \in \mathcal{J}_m} D_q(\iota_* \mu_m \| \pi_m) + O_\alpha(1), \qquad q \in [0,\infty],$$

and the additive error is in fact bounded between 0 and $2 \log \varphi$.

Proof. Theorem 5.11 gives

$$\inf_{\iota \in \mathcal{J}_m} D_q(\iota_* \mu_m \| \pi_m) = (m-1) \log \varphi - \log \pi(0) - H_q(\mu_m).$$

Now apply Proposition 5.13 to substitute $H_q(\mu_m) = \log m + O_\alpha(1)$, uniformly in q. The final claim is the last part of Theorem 5.11. □

Theorem 5.16 (Universal lossless coarse-graining envelope)**.** *Assume $\beta = \alpha$ and $m \geq 6$. Let $S_m(\alpha)$ be the realized Sturmian language and μ_m its factor law. Define the class of deterministic maps that are injective on the realized source:*

$$\mathcal{G}_m := \{G_m : \Omega_m \to X_m \ \textit{deterministic} : \ G_m|_{S_m(\alpha)} \ \textit{injective}\}.$$

For $G_m \in \mathcal{G}_m$, write $\nu_m^G := (G_m)_* \mu_m$. Then for every $q \in [0,\infty]$,

$$\inf_{G_m \in \mathcal{G}_m} D_q(\nu_m^G \| \pi_m) = (m-1) \log \varphi - \log \pi(0) - H_q(\mu_m),$$

and a map G_m attains the infimum if and only if $G_m(S_m(\alpha)) \subset X_m^{00}$. Moreover, the canonical fold satisfies

$$0 \leq D_q(\pi_m^{\mathrm{fold}} \| \pi_m) - \inf_{G_m \in \mathcal{G}_m} D_q(\nu_m^G \| \pi_m) \leq 2 \log \varphi.$$

Proof. For any $G_m \in \mathcal{G}_m$, the restriction $\iota := G_m|_{S_m(\alpha)} : S_m(\alpha) \hookrightarrow X_m$ is injective, and $\nu_m^G = \iota_* \mu_m$. Thus the optimization over $\mathcal{G}_m$ reduces to the injective relabeling problem of Theorem 5.11. The infimum, the characterization of optimizers, and the bounded gap for the canonical fold all follow from that theorem. □

5.4 Primitive SFT Asymptotic Extension

Theorem 5.17 (Primitive SFT asymptotic Renyi barrier)**.** *Let Σ_A be a primitive one-step shift of finite type with topological entropy $h = \log \lambda$. Write X_m^A for its admissible words of length m, and let π_m^A be the m-block law of its Parry measure. Then there exist constants $c_A, C_A > 0$ such that*

$$c_A e^{-mh} \le \pi_m^A(w) \le C_A e^{-mh} \qquad (w \in X_m^A).$$

Consequently, if $(\nu_m)_{m\ge1}$ is any family with $\nu_m \in \Delta(X_m^A)$ and

$$\limsup_{m\to\infty} \frac{1}{m} \log |\operatorname{supp}(\nu_m)| \le \gamma < h,$$

then

$$\liminf_{m\to\infty} \frac{1}{m} D_q(\nu_m \| \pi_m^A) \ge h - \gamma \qquad (q \in [0,\infty]).$$

Conversely, for every $\gamma \in [0,h]$, there exists a family $(\nu_m)_{m\ge1}$ with $\nu_m \in \Delta(X_m^A)$ such that

$$\frac{1}{m} \log |\operatorname{supp}(\nu_m)| \to \gamma, \qquad \frac{1}{m} D_q(\nu_m \| \pi_m^A) \to h - \gamma \qquad (q \in [0,\infty]).$$

Proof. Let u and v be positive left and right Perron-Frobenius eigenvectors for A, normalized so that $\sum_i u_i v_i = 1$. The standard Parry formula for primitive one-step shifts [**?**, **?**] gives

$$\pi_m^A(i_0 \cdots i_{m-1}) = u_{i_0} v_{i_{m-1}} \lambda^{-(m-1)}$$

for every admissible word $i_0 \cdots i_{m-1} \in X_m^A$. Since the alphabet is finite and $u_i v_j > 0$ on admissible endpoint pairs, there are constants $c_A, C_A > 0$ such that

$$c_A e^{-mh} \le \pi_m^A(w) \le C_A e^{-mh} \qquad (w \in X_m^A).$$

Now let $S_m := \operatorname{supp}(\nu_m)$. By Proposition 5.7,

$$D_q(\nu_m \| \pi_m^A) \ge -\log \pi_m^A(S_m) \qquad (q \in [0,\infty]).$$

Using the upper block-mass bound,

$$\pi_m^A(S_m) \le C_A |S_m| e^{-mh},$$

so

$$\frac{1}{m} D_q(\nu_m \| \pi_m^A) \ge h - \frac{1}{m} \log |S_m| - \frac{\log C_A}{m}.$$

Taking $\liminf$ proves the lower bound.

For sharpness, first fix $\gamma \in [0,h)$ and set $k_m := \max\{1, \lfloor e^{\gamma m} \rfloor\}$. Choose any subset $S_m \subset X_m^A$ with $|S_m| = k_m$, and define $\nu_m := \pi_m^A(\cdot \mid S_m)$. Then

$$c_A k_m e^{-mh} \le \pi_m^A(S_m) \le C_A k_m e^{-mh},$$

and Proposition 5.7 yields

$$D_q(\nu_m \| \pi_m^A) = -\log \pi_m^A(S_m) = mh - \log k_m + O_A(1) \qquad (q \in [0,\infty]).$$

Since $m^{-1} \log k_m \to \gamma$, this gives the desired limit. For $\gamma = h$, take $\nu_m := \pi_m^A$; then every $D_q(\nu_m \| \pi_m^A)$ is zero, while the language growth of a primitive shift satisfies $|X_m^A| = e^{hm + o(m)}$ [**?**], so $m^{-1} \log |\operatorname{supp}(\nu_m)| \to h$. □

Remark 5.18 (Scope of the primitive-SFT extension). Theorem 5.17 is intentionally asymptotic. It uses only the comparability $\pi_m^A(w) \asymp e^{-mh}$ of Parry block masses on a primitive shift. Unlike the golden-mean results of Theorem 5.9, it does not provide a finite-m mass envelope $M_m(k)$, exact support-constrained optimizers, or a closed formula beyond the exponential-rate slope.

Theorem 5.19 (Primitive SFT injective placement barrier)**.** *Let Σ_A be a primitive one-step shift of finite type with topological entropy $h = \log \lambda$. Let u, v be positive left and right Perron-Frobenius eigenvectors for A, normalized so that $\sum_i u_i v_i = 1$. For an admissible word $w = i_0 \cdots i_{m-1}$, the Parry formula gives $\pi_m^A(w) = u_{i_0} v_{i_{m-1}} \lambda^{-(m-1)}$. Define*

$$c_* := \max_{i,j} u_i v_j,$$

and let (i_*, j_*) attain this maximum. Write

$$\mathcal{C}_m(i_*, j_*) := \{w \in X_m^A : w_1 = i_*,\ w_m = j_*\}.$$

Then there exist $\kappa_A > 0$ and m_0 such that for all $m \geq m_0$,

$$|\mathcal{C}_m(i_*, j_*)| \geq \kappa_A e^{mh}.$$

Moreover, if μ_m is any probability vector on a finite set S_m, then for every injection $\iota : S_m \hookrightarrow X_m^A$ and every $q \in [0, \infty]$,

$$D_q(\iota_* \mu_m \| \pi_m^A) \geq (m-1)h - \log c_* - H_q(\mu_m).$$

If $|S_m| \leq |\mathcal{C}_m(i_*, j_*)|$, the lower bound is attained by any injection into $\mathcal{C}_m(i_*, j_*)$, and

$$\inf_{\iota : S_m \hookrightarrow X_m^A} D_q(\iota_* \mu_m \| \pi_m^A) = (m-1)h - \log c_* - H_q(\mu_m).$$

In particular, if $\log |S_m| = o(m)$, then

$$\inf_{\iota : S_m \hookrightarrow X_m^A} D_q(\iota_* \mu_m \| \pi_m^A) = mh - H_q(\mu_m) + O_A(1),$$

where the $O_A(1)$ term is uniform in q.

Proof. By the Perron-Frobenius theorem, $(A^{m-1})_{i_* j_*} = \lambda^{m-1}(v_{i_*} u_{j_*} + o(1))$, so there exist $\kappa_A > 0$ and m_0 with $|\mathcal{C}_m(i_*, j_*)| = (A^{m-1})_{i_* j_*} \geq \kappa_A \lambda^{m-1}$ for $m \geq m_0$.

Fix an injection ι and write $p_s = \mu_m(s)$, $\rho_s = \pi_m^A(\iota(s))$. By definition of c_*, $\rho_s \leq c_* \lambda^{-(m-1)}$ for all $s \in S_m$.

For $q = 1$: $D_1 = \sum_s p_s \log(p_s/\rho_s) \geq \sum_s p_s \log(p_s/(c_* \lambda^{-(m-1)})) = (m-1)h - \log c_* - H_1(\mu_m)$.

For $q \in (0, \infty) \setminus \{1\}$: the bound $\rho_s \leq c_* \lambda^{-(m-1)}$ gives $\rho_s^{1-q} \geq (c_* \lambda^{-(m-1)})^{1-q}$ when $q > 1$ and $\rho_s^{1-q} \leq (c_* \lambda^{-(m-1)})^{1-q}$ when $q < 1$. In both cases the prefactor $(q-1)^{-1}$ produces the same lower bound $D_q \geq (m-1)h - \log c_* - H_q(\mu_m)$.

For $q = \infty$: $D_\infty = \log \max_s (p_s/\rho_s) \geq (m-1)h - \log c_* - H_\infty(\mu_m)$.

For $q = 0$: $\pi_m^A(\iota(S_m)) \leq |S_m| c_* \lambda^{-(m-1)}$, so $D_0 = -\log \pi_m^A(\iota(S_m)) \geq (m-1)h - \log c_* - H_0(\mu_m)$.

For attainability, if $|S_m| \leq |\mathcal{C}_m(i_*, j_*)|$, inject S_m into $\mathcal{C}_m(i_*, j_*)$. All image words then have identical Parry mass $c_* \lambda^{-(m-1)}$, so the lower bound is attained for every q. If $\log |S_m| = o(m)$, then $|S_m| \leq \kappa_A \lambda^{m-1}$ for large m, and absorbing $(m-1)h - \log c_*$ into $mh + O_A(1)$ yields the last claim. $\square$

Corollary 5.20 (Low-complexity source barrier)**.** *Let Σ_A be a primitive one-step SFT with topological entropy $h = \log \lambda$ and Parry m-block law π_m^A. Let $(S_m, \mu_m)_{m\geq 1}$ be a family of finite probability spaces satisfying $|S_m| = e^{o(m)}$ and $H_q(\mu_m) = \log |S_m| + O(1)$ uniformly in $q \in [0, \infty]$. Then for all sufficiently large m, any injection $\iota_m : S_m \hookrightarrow X_m^A$ satisfies*

$$D_q(\iota_{m*}\mu_m \| \pi_m^A) = mh - \log |S_m| + O(1), \qquad q \in [0, \infty],$$

with constants uniform in q. In particular, if $|S_m| = m + O(1)$, then

$$D_q(\iota_{m*}\mu_m \| \pi_m^A) = mh - \log m + O(1).$$

Proof. By Theorem 5.19, for large m the infimum over injections equals $(m-1)h - \log c_* - H_q(\mu_m)$. Substituting $H_q(\mu_m) = \log |S_m| + O(1)$ and absorbing $(m-1)h - \log c_*$ into $mh + O_A(1)$ gives the first claim. The lower bound is attained since $|S_m| = e^{o(m)}$ eventually fits inside the heaviest endpoint class. The linear-complexity case follows from $\log |S_m| = \log m + O(1)$. □

Remark 5.21 (Reading the numerical section)*.* The numerical examples are organized around a simple two-stage audit. First, discrepancy estimates test whether the empirical histogram is already close to its deterministic folded law. Second, comparison with the Parry baseline tests whether that folded law is compatible with the ambient symbolic state space or instead carries a structural support deficit that no increase in sample size can remove. The purpose of Section 6 is to illustrate this separation in a completely explicit benchmark, not to survey a large family of unrelated fold maps.

6 Numerical Illustrations

This section complements the theory rather than attempting an exhaustive parameter study. We begin with the canonical slice $\beta = \alpha$, where the results of Sections 4 and 5 are sharp, and then turn to additional parameter families and an i.i.d. baseline. In every case, empirical agreement with the folded limit law improves with sample size, whereas disagreement with the Parry baseline can remain large because it reflects a structural mismatch. In the canonical slice, Theorem 4.10 rules out information loss along the fold, and Theorem 5.11 identifies the best injective placement of the realized Sturmian law in X_m.

6.1 Numerical Setup

Experiment family	Parameter grid
Rotation sweep	$m \in \{6, 8, 10, 12, 14, 16, 18\}$, $N \in \{10^4, 3 \times 10^4, 10^5, 3 \times 10^5\}$, $\beta \in \{0.2, 0.3, 0.5, 0.6180339887\}$, $x_0 \in \{0.123, 0.271, 0.731\}$
Rotation families	golden, metallic means $A \in \{2, 3, 5\}$, and large first partial quotient families $[0; B, 1, 1, \dots]$ with $B \in \{10, 30, 100\}$
IID block baselines	$m \in \{6, 8, 10, 12, 14, 16, 18\}$, $N \in \{2000, 5000, 10^4, 3 \times 10^4, 10^5\}$, seeds $\{1, 2, 3, 4, 5\}$, confidence level 95%

Table 1: Parameter ranges used in the numerical examples.

The reported tables are extracted from this fixed parameter grid rather than from isolated hand-picked samples. For the rotation experiments, the same orbit sample is used to compute the

empirical microstate law, the folded histogram, the discrepancy certificate, and the cross-resolution residual. The overlines in Table 4 denote averages over the sampled orbit phases listed in Table 1, and Table 5 further averages over the reported β-grid. The large value $N = 3 \times 10^5$ in the residual and Parry tables is intentional: it drives the discrepancy certificate into the 10^{-5} regime, so the remaining signal is forced to come from structural effects rather than from sampling noise alone.

All slope families in Table 1 are bounded type. The family $[0; B, 1, 1, \dots]$ is included as a resonant stress test with a large first partial quotient, not as a genuinely unbounded-type example. At the level of reproducibility, the manuscript includes the exact emitted tables used in the PDF through the `generated/` files, with fixed orbit phases and IID seeds specified by Table 1; a short reproducibility note accompanying the source bundle records the generating scripts and table provenance.

6.2 Canonical Slice

Theorem 4.10 is an exact structural statement, so the main quantitative verification target in this section is Theorem 5.11. We therefore start with a canonical-slice table that contains no averaging over β or x_0 and is computed from the exact Lebesgue factor law rather than from a finite orbit sample. The table begins at $m = 6$, the first resolution at which the heaviest endpoint class X_m^{00} already has enough capacity to realize the injective optimum exactly.

m	exact $L_1(m)$	exact $\mathrm{KL}(\pi_m^{\mathrm{fold}} \| \pi_m)$	gap	exact D_{TV}
6	0.832	1.199	0.368	0.668
8	1.540	1.908	0.368	0.839
10	2.308	2.676	0.368	0.920
12	3.077	3.444	0.368	0.965
14	3.919	4.287	0.368	0.985
16	4.762	5.129	0.368	0.994
18	5.604	5.972	0.368	0.997

Table 2: Canonical Sturmian slice $\beta = \alpha$: exact quantities computed from the Lebesgue factor law, with no averaging over β or x_0. Here $L_1(m)$ is the optimal injective floor from Theorem 5.11, and the constant gap verifies that the canonical folded law stays within bounded additive distance of the best injective relabeling while remaining far from the Parry baseline.

Table 2 directly verifies the sharp finite-m content of Theorem 5.11. The optimal injective floor $L_1(m)$ and the exact folded-to-Parry divergence $\mathrm{KL}(\pi_m^{\mathrm{fold}} \| \pi_m)$ both grow on the scale $m \log \varphi - \log m$, while their difference is the exact constant

$$\mathrm{KL}(\pi_m^{\mathrm{fold}} \| \pi_m) - L_1(m) = (1 - \varphi^{-3}) \log \varphi = 0.3676131227688\dots$$

for every displayed row. This follows from the exact identity in Theorem 5.11 together with the evaluation $\eta_{11}(\pi_m^{\mathrm{fold}}) - \eta_{00}(\pi_m^{\mathrm{fold}}) = -\varphi^{-3}$ on the canonical Sturmian slice. At the finite resolutions shown here, the canonical folded law already lies well below the theorem's universal upper bound $2 \log \varphi$ above the injective optimum. This table therefore provides the structural baseline for the rest of the section.

6.3 Finite-Sample Validation

m	N	D_{TV} (fold)	D_{KL} (fold)	TV upper bd	KL upper bd
8	10000	0.000126	6.01e-08	0.0216	0.0167
12	30000	2.62e-05	1.71e-09	0.0117	0.00735
16	100000	1.74e-05	1.04e-09	0.00493	0.00174
18	300000	1.32e-05	4.52e-10	0.00196	0.00031

Table 3: Golden-slope rotation scan with the canonical window $\beta = \alpha$: folded empirical TV and relative entropy against the Lebesgue pushforward target, together with the explicit upper bounds from Section 4.

Table 3 is reported for the golden slope with the canonical window $\beta = \alpha$. Across the parameter range shown here, the observed folded relative entropy remains well below the explicit upper bound. The constants are conservative, but the observed dependence on (m, N) follows the predicted scale. The same canonical slice is also the regime in which Theorem 5.11, Corollary 5.15, and Table 2 predict a very different behavior relative to the Parry baseline: increasing the sample size improves agreement with π_m^{fold}, but it does not remove the structural gap to π_m. Once the empirical histogram is close to π_m^{fold}, the injective floor from Table 2 becomes the relevant benchmark for the remaining Parry mismatch.

6.4 Cross-Resolution Residuals

Table 4: Representative cross-resolution residuals $E_m = D_{\mathrm{TV}}(\hat{\pi}_{m,N}, (\rho_{m+1\to m})_* \hat{\pi}_{m+1,N})$ for three irrational families, together with the mean star discrepancy $\overline{D_N^*}$.

m	N	$\overline{D_N^*}$	golden $\overline{E_m}$	metal $A = 2$ $\overline{E_m}$	resonant $B = 100$ $\overline{E_m}$
6	300000	1.01e-05	0.177	0.156	0.365
8	300000	1.01e-05	0.22	0.185	0.365
10	300000	1.01e-05	0.22	0.195	0.365
12	300000	1.01e-05	0.22	0.195	0.365

Table 4 makes the separation between finite-sample and structural effects concrete. At $N = 3 \times 10^5$, the mean star discrepancy in the reported golden slice is already of order 10^{-5}, whereas the projective residual remains of order 10^{-1}. For the resonant family $[0; 100, 1, 1, \dots]$, the residual is larger still. This is exactly the phenomenon captured by Theorem 3.4: small orbit discrepancy does not force small cross-resolution residual, because the residual is driven by the frequency with which a final 1 is appended to prefixes whose Fibonacci value has already crossed the threshold F_{m+1}.

Corollary 3.6 gives a quantitative bound for this phenomenon: the empirical residual is controlled by an explicit discrepancy term together with the structural TV gaps to the Parry baselines at levels m and $m + 1$. In particular, once the discrepancy term is much smaller than those structural terms, further increases in N cannot force the residual onto the discrepancy scale.

The table is intended as a compressed visualization of that boundary law. The boundary event itself is binary and sparse, so plotting raw event traces would add little beyond the residual summary; the summary is the observable obtained after folding and truncation, and it is precisely this quantity that the theorem identifies.

6.5 Additional Parameter Choices

Table 5: Off-canonical robustness summary at $N = 3 \times 10^5$, averaged over the β and x_0 grid from Table 1. Unlike Table 2, this table is not the canonical-slice theorem verification; it is retained only to show that the same qualitative TV and KL mismatch to π_m remains visible across bounded-type and resonant families away from the exact slice $\beta = \alpha$. The resonant family is $[0; 100, 1, 1, \dots]$.

m	golden D_{TV}	golden D_{KL}	metal $A = 2$ D_{TV}	metal $A = 2$ D_{KL}	resonant D_{TV}	resonant D_{KL}
6	0.506	0.804	0.477	0.705	0.790	1.76
8	0.729	1.48	0.701	1.39	0.831	2.54
10	0.868	2.23	0.857	2.12	0.853	3.32
12	0.941	3.01	0.934	2.87	0.931	4.10
14	0.973	3.84	0.970	3.68	0.968	4.89
16	0.988	4.67	0.987	4.52	0.986	5.68
18	0.995	5.52	0.994	5.39	0.994	6.48

Table 5 is intentionally secondary to Table 2. It is kept not as the main verification of Section 5, but as a broader robustness illustration away from the exact canonical slice. At the fixed large sample size $N = 3 \times 10^5$, the golden discrepancy upper bound is already only 9.33×10^{-5}, yet the total-variation and relative-entropy gaps to the Parry measure still grow rapidly with m across both bounded-type and resonant families. In particular, for the golden branch the total variation distance rises from 0.506 at $m = 6$ to 0.995 at $m = 18$. This is the same qualitative regime predicted by Corollary 5.1, Corollary 5.15, and the exact support-budget formulas of Section 5: once m grows, the dominant contribution is structural mismatch rather than sampling error.

The comparison across families is also useful conceptually. The metallic and resonant examples differ in constants, but they exhibit the same qualitative obstruction: after folding into X_m, low effective complexity prevents convergence to the maximal-entropy baseline. For that reason the paper does not try to benchmark many alternative local normalization rules numerically. By the support-budget law of Section 5, any alternative fold that still produces a subexponential-support family on X_m must retain a positive Parry entropy density.

6.6 Comparison with IID Baselines

Table 6: IID block baselines: deviation of the folded histogram from the Parry baseline together with 95% concentration radii and bounds, averaged over seeds.

source	D_{TV}	gap	95% TV rad.	95% TV bd.	$D_{\mathrm{KL}}(\hat{p}\|p_{\mathrm{true}})$	95% KL rad.
Bernoulli blocks	0.161	0.159	0.0414	0.201	0.00215	0.0496
Parry blocks	0.0261	0	0.0414	0.0414	0.00212	0.0496

The IID comparison is useful because it displays the decomposition in Corollary 5.1 in two opposite regimes. When the data-generating law is matched to the reference baseline, the discrepancy term controls the observed error. For the folded rotation data, by contrast, the structural term persists and eventually dominates. No systematic comparison with alternative folds is included here. The purpose of the numerical section is to corroborate the canonical theory developed in Sections 2–5, and the support-barrier results already show what any such comparison could and could not change: a different fold may alter constants or the effective support exponent, but it cannot erase the Parry penalty while remaining inside the same low-complexity regime.

For reference, the standard high-probability envelopes on a finite alphabet E with $|E| = F_{m+2}$ are

$$\mathrm{TV}(\hat{p}_N, p) \lesssim \sqrt{\frac{\log(2^{F_{m+2}}/\delta)}{2N}}, \qquad \mathrm{KL}(\hat{p}_N \| p) \lesssim \frac{F_{m+2}\log(N+1) + \log(1/\delta)}{N}$$

at confidence level $1-\delta$; see [**?**, **?**]. These apply uniformly to both the IID and rotation settings. When $p = \pi_m$ and the data are Parry-distributed, the observed errors in the table fall well within this envelope. When $p = \pi_m$ but the data come from folded rotation orbits, the structural gap dominates.

6.7 A Nonlinear Circle-Map Realization

The benchmark is not confined to rigid rotations as state equations. The following theorem shows that the entire audit is a conjugacy invariant.

Theorem 6.1 (Audit invariance under topological conjugacy)**.** *Let $f : S^1 \to S^1$ be an orientation-preserving circle homeomorphism topologically conjugate to the irrational rotation $R_\alpha : x \mapsto x + \alpha \pmod 1$ via an orientation-preserving homeomorphism h satisfying $h \circ f = R_\alpha \circ h$. Fix an anchor $a \in S^1$ with $h(a) = 0$, and define the coding window $I_a := [a, f(a))$. For $x \in S^1$, let*

$$s_t^{(f)}(x) := \mathbf{1}_{I_a}(f^t(x)), \qquad s_t^{(\alpha)}(y) := \mathbf{1}_{[0,\alpha)}(y + t\alpha \bmod 1).$$

Then for every $x \in S^1$ and $y := h(x)$,

$$s_t^{(f)}(x) = s_t^{(\alpha)}(y) \qquad (t \geq 0).$$

Consequently, for every $m, N \geq 1$, the length-m microstate laws, folded laws, empirical histograms, discrepancy certificates, Parry comparisons, higher-block conjugacy, and all Renyi/KL conclusions of Sections 3–5 are identical for the two systems. In particular, if f is a C^2 orientation-preserving circle diffeomorphism with irrational rotation number α, the Denjoy theorem provides the required conjugacy, and the full two-stage audit applies.

Proof. Since h is orientation-preserving, the image of $I_a = [a, f(a))$ is the positively oriented arc from $h(a) = 0$ to $h(f(a)) = R_\alpha(h(a)) = \alpha$, i.e., $h(I_a) = [0, \alpha)$. For any $x \in S^1$ with $y = h(x)$, the conjugacy gives $h(f^t(x)) = R_\alpha^t(y) = y + t\alpha \pmod 1$, so

$$s_t^{(f)}(x) = \mathbf{1}_{I_a}(f^t(x)) = \mathbf{1}_{h(I_a)}(h(f^t(x))) = \mathbf{1}_{[0,\alpha)}(y + t\alpha \bmod 1) = s_t^{(\alpha)}(y).$$

Since the symbol sequences agree pointwise for all t, the length-m sliding windows, and therefore all empirical counts, folded histograms, and derived quantities, are identical. □

As a concrete illustration, consider the standard circle map

$$f_{K,\Omega}(x) = x + \Omega - \frac{K}{2\pi}\sin(2\pi x) \pmod 1$$

with $K = 0.3$ and Ω tuned by bisection on the lift so that the rotation number equals φ^{-1}, giving $\Omega \approx 0.6166931270470741$. With anchor $a = 0.2$, coding window $I = [a, f(a))$, and initial point $x_0 = a$, the resulting symbolic sequence is the canonical golden Sturmian coding in conjugacy coordinates. At $m = 12$ and $N = 10^5$, the empirical folded histogram satisfies

$$\mathrm{TV}(\hat{\pi}_{12,N}, \pi_{12}^{\mathrm{fold}}) = 2.43 \times 10^{-5}, \qquad \mathrm{KL}(\hat{\pi}_{12,N} \| \pi_{12}^{\mathrm{fold}}) = 2.95 \times 10^{-9},$$

while the gaps to the Parry baseline remain

$$\mathrm{TV}(\hat{\pi}_{12,N}, \pi_{12}) = 0.9650, \qquad \mathrm{KL}(\hat{\pi}_{12,N} \| \pi_{12}) = 3.4443.$$

The first-stage certificate is already at the 10^{-5} level, while the second-stage structural mismatch persists at order 10^0. The two-stage audit thus functions identically on a genuinely nonlinear state-space model.

7 Concluding Remarks

The present benchmark is not valuable merely because it is fully computable. Its stronger role is to calibrate the universal injective-placement barrier for lossless coarse-grainings into primitive shifts of finite type. In general, any deterministic coarse-graining that is injective on the realized source retains the order-uniform divergence rate $D_q^{\mathrm{opt}} = mh - H_q(\mu_m) + O(1)$ (Theorem 5.19). The golden-mean benchmark is exceptional because this universal envelope becomes finite-m exact (Theorem 5.16), the optimizer is explicit, and the actual canonical fold remains within a bounded additive distance of that optimizer while still admitting a nonlinear circle-map realization.

The framework separates two questions: whether a folded histogram is already statistically close to its deterministic limit law, and whether that limit law is compatible with a chosen ambient equilibrium. In the present example the two effects can be separated analytically. Discrepancy estimates give finite-sample control, while the comparison with the Parry measure identifies the part of the mismatch that persists even in the large-sample limit.

The canonical Sturmian slice is the point at which the picture becomes especially rigid. There the fold is not only injective on realized blocks; it is conjugate to the classical higher-block presentation of the Sturmian source. At the same time, the endpoint structure of the golden-mean Parry cylinders is simple enough to make the Parry comparison exact. Theorem 5.11 shows that even the best injective placement of the realized Sturmian law in X_m still has divergence $m \log \varphi - \log m + O_\alpha(1)$ for bounded-type slopes, uniformly in the Renyi order. The folded law lies within a bounded additive distance of this optimum. By topological conjugacy, the entire audit transfers verbatim to C^2 circle diffeomorphisms with irrational rotation number (Section 6.7).

The limitations are also clear. The exact finite-m formulas rely on the special geometry of the golden-mean shift, and the primitive-SFT extension in Section 5 is only asymptotic. For that reason the paper is best viewed as a benchmark that calibrates a general theory rather than as the general theory itself. The numerical section is designed to verify the sharp statements available in the canonical slice and to illustrate the same mechanism in nearby parameter families and on a nonlinear circle map, not to claim a general computational survey of fold maps.

Appendix B provides the local normalization description of the fold, while Appendix C shows that the Fibonacci weighting is the unique choice satisfying a natural contiguous numeration axiom. These appendices, together with the microstate lift identities of Appendix A, do not change the main conclusion: in this benchmark, sampling error, support deficit, and within-fiber degeneracy can be distinguished analytically, and the dominant mismatch with the Parry baseline comes from the source/reference pairing rather than from the fold itself.

A Fiber-Uniform Lifts and the Parry Lift

This appendix collects the microstate-level lift identities associated with the canonical fold. They are used only to interpret the hidden degeneracy term in Corollary 4.11 and may be skipped on a first reading.

Definition A.1 (Fiber multiplicity and uniform lift)**.** For $x \in X_m$, define the fiber multiplicity

$$d_m(x) := \left|\mathrm{Fold}_m^{-1}(x)\right|.$$

For $\eta \in \Delta(X_m)$, define its fiber-uniform lift

$$(Q_m\eta)(a) := \frac{\eta(\mathrm{Fold}_m(a))}{d_m(\mathrm{Fold}_m(a))}, \qquad a \in \Omega_m.$$

Then $(\mathrm{Fold}_m)_*(Q_m\eta) = \eta$. For $\mu \in \Delta(\Omega_m)$, let

$$\pi := (\mathrm{Fold}_m)_*\mu \in \Delta(X_m), \qquad \mu^e := Q_m\pi.$$

Proposition A.2 (Exact entropy ledger for the fold quotient)**.** *For every $\mu \in \Delta(\Omega_m)$, with $\pi = (\mathrm{Fold}_m)_*\mu$ and $\mu^e = Q_m\pi$,*

$$H(\mu) = H(\pi) + \sum_{x \in X_m} \pi(x) \log d_m(x) - \mathrm{KL}(\mu \| \mu^e).$$

Equivalently,

$$H(\mu) - H(\pi) = \mathbb{E}_\pi[\log d_m(X)] - \mathrm{KL}(\mu \| \mu^e).$$

Proof. Write $x_a := \mathrm{Fold}_m(a)$. By Definition A.1,

$$\mu^e(a) = \frac{\pi(x_a)}{d_m(x_a)}.$$

Hence

$$\begin{aligned}\mathrm{KL}(\mu \| \mu^e) &= \sum_{a \in \Omega_m} \mu(a) \log \frac{\mu(a)}{\mu^e(a)} \\ &= -H(\mu) - \sum_{a \in \Omega_m} \mu(a) \log \pi(x_a) + \sum_{a \in \Omega_m} \mu(a) \log d_m(x_a).\end{aligned}$$

Since $\sum_{a:\, x_a = x} \mu(a) = \pi(x)$, the two sums become

$$\sum_{a \in \Omega_m} \mu(a) \log \pi(x_a) = \sum_{x \in X_m} \pi(x) \log \pi(x) = -H(\pi)$$

and

$$\sum_{a \in \Omega_m} \mu(a) \log d_m(x_a) = \sum_{x \in X_m} \pi(x) \log d_m(x).$$

Rearranging gives the claim. □

Proposition A.3 (KL isometry of fiber-uniform lifts)**.** *For every $\pi, \eta \in \Delta(X_m)$,*

$$\mathrm{KL}(Q_m\pi \| Q_m\eta) = \mathrm{KL}(\pi \| \eta).$$

Proof. Again write $x_a := \mathrm{Fold}_m(a)$. Then

$$\frac{(Q_m\pi)(a)}{(Q_m\eta)(a)} = \frac{\pi(x_a)}{\eta(x_a)},$$

so

$$\begin{aligned}\mathrm{KL}(Q_m\pi\|Q_m\eta) &= \sum_{a\in\Omega_m}\frac{\pi(x_a)}{d_m(x_a)}\log\frac{\pi(x_a)}{\eta(x_a)}\\ &= \sum_{x\in X_m}\left(\sum_{a:\,x_a=x}\frac{\pi(x)}{d_m(x)}\right)\log\frac{\pi(x)}{\eta(x)} = \mathrm{KL}(\pi\|\eta).\end{aligned}$$

□

Theorem A.4 (Pythagorean decomposition for the canonical fold)**.** *Let $\mu \in \Delta(\Omega_m)$, let $\pi = (\mathrm{Fold}_m)_*\mu$, and let $\mu^e = Q_m\pi$. Then for every $\eta \in \Delta(X_m)$,*

$$\mathrm{KL}(\mu\|Q_m\eta) = \mathrm{KL}(\mu\|\mu^e) + \mathrm{KL}(\pi\|\eta).$$

Consequently, μ^e is the unique I-projection of μ onto the fiber-uniform model family

$$\mathcal{E}_m := \{Q_m\eta : \eta \in \Delta(X_m)\} \subset \Delta(\Omega_m),$$

and

$$\inf_{\eta\in\Delta(X_m)} \mathrm{KL}(\mu\|Q_m\eta) = \mathrm{KL}(\mu\|\mu^e).$$

Proof. Let $x_a := \mathrm{Fold}_m(a)$. Since

$$\mu^e(a) = \frac{\pi(x_a)}{d_m(x_a)}, \qquad (Q_m\eta)(a) = \frac{\eta(x_a)}{d_m(x_a)},$$

one has

$$\log\frac{\mu^e(a)}{(Q_m\eta)(a)} = \log\frac{\pi(x_a)}{\eta(x_a)}.$$

Therefore

$$\begin{aligned}\mathrm{KL}(\mu\|Q_m\eta) - \mathrm{KL}(\mu\|\mu^e) &= \sum_{a\in\Omega_m}\mu(a)\log\frac{\mu^e(a)}{(Q_m\eta)(a)}\\ &= \sum_{a\in\Omega_m}\mu(a)\log\frac{\pi(x_a)}{\eta(x_a)}\\ &= \sum_{x\in X_m}\left(\sum_{a:\,x_a=x}\mu(a)\right)\log\frac{\pi(x)}{\eta(x)} = \mathrm{KL}(\pi\|\eta).\end{aligned}$$

This proves the decomposition. Since $\mathrm{KL}(\pi\|\eta) \geq 0$, with equality if and only if $\eta = \pi$, the minimizer is unique and equals $\mu^e = Q_m\pi$. □

Proposition A.5 (Dual characterization of the fiber-uniform lift)**.** *For any $\pi \in \Delta(X_m)$, define the linear family*

$$\mathcal{L}_\pi := \{\nu \in \Delta(\Omega_m) : (\mathrm{Fold}_m)_*\nu = \pi\}.$$

Then $Q_m\pi$ is the unique maximum-entropy element of $\mathcal{L}_\pi$:

$$H(\nu) \leq H(Q_m\pi) \qquad (\nu \in \mathcal{L}_\pi),$$

with equality if and only if $\nu = Q_m\pi$. Hence, for any $\mu \in \Delta(\Omega_m)$ with $\pi = (\mathrm{Fold}_m)_*\mu$, the lift $Q_m\pi$ is simultaneously:

1. the maximum-entropy projection onto $\mathcal{L}_\pi$, and

2. the I-projection of μ onto the fiber-uniform family $\mathcal{E}_m = \{Q_m\eta : \eta \in \Delta(X_m)\}$.

Proof. For any $\nu \in \mathcal{L}_\pi$, Proposition A.2 gives

$$H(\nu) = H(\pi) + \sum_{x \in X_m} \pi(x) \log d_m(x) - \mathrm{KL}(\nu \| Q_m\pi).$$

The first two terms are independent of ν, and $\mathrm{KL}(\nu\|Q_m\pi) \geq 0$, so $H(\nu) \leq H(Q_m\pi)$ with equality if and only if $\nu = Q_m\pi$. The I-projection statement is Theorem A.4. □

Remark A.6 (Relation to classical information projection). Theorem A.4 and Proposition A.5 together realize the classical duality of Csiszár [**?**] in the present finite-state setting: the sufficient statistic is the folded observable Fold_m, and the fiber-uniform family $\mathcal{E}_m$ is exactly the family of laws that are uniform within each fiber of this coarse-graining. The maximum-entropy characterization and the I-projection characterization identify the same object $Q_m\pi$ from complementary perspectives.

Corollary A.7 (Microstate decomposition relative to the Parry lift)**.** *Let $\mu \in \Delta(\Omega_m)$, let $\pi = (\mathrm{Fold}_m)_*\mu$, and let $\mu^e = Q_m\pi$. Then*

$$\mathrm{KL}(\mu\|Q_m\pi_m) = \mathrm{KL}(\mu\|\mu^e) + \mathrm{KL}(\pi\|\pi_m).$$

If $|\operatorname{supp}(\pi)| \leq k$, then

$$\mathrm{KL}(\mu\|Q_m\pi_m) \geq \mathrm{KL}(\mu\|\mu^e) - \log M_m(k).$$

In particular, assume $\beta = \alpha$ and $m \geq 6$. Then

$$\mathrm{KL}(\mu_m\|Q_m\pi_m) = \mathbb{E}_{\pi_m^{\mathrm{fold}}}\big[\log d_m(Y)\big] + \mathrm{KL}(\pi_m^{\mathrm{fold}}\|\pi_m),$$

$$\mathrm{KL}(\mu_m\|Q_m\pi_m) \geq \mathbb{E}_{\pi_m^{\mathrm{fold}}}\big[\log d_m(Y)\big] + (m-1)\log\varphi - \log\pi(0) - \log(m+1),$$

and for every $N \geq 1$,

$$\mathrm{KL}(\hat{\mu}_{m,N}\|Q_m\pi_m) = \mathbb{E}_{\hat{\pi}_{m,N}}\big[\log d_m(Y)\big] + \mathrm{KL}(\hat{\pi}_{m,N}\|\pi_m),$$

$$\mathrm{KL}(\hat{\mu}_{m,N}\|Q_m\pi_m) \geq \mathbb{E}_{\hat{\pi}_{m,N}}\big[\log d_m(Y)\big] + (m-1)\log\varphi - \log\pi(0) - \log(m+1).$$

Proof. The first identity is Theorem A.4 with $\eta = \pi_m$. If $|\operatorname{supp}(\pi)| \leq k$, then Theorem 5.9 gives $\mathrm{KL}(\pi\|\pi_m) \geq -\log M_m(k)$, which yields the second inequality.

Under $\beta = \alpha$, Corollary 4.11 identifies $\mathrm{KL}(\mu_m\|\mu_m^e)$ and $\mathrm{KL}(\hat{\mu}_{m,N}\|\hat{\mu}_{m,N}^e)$ with the corresponding degeneracy averages. The folded laws π_m^{fold} and $\hat{\pi}_{m,N}$ are supported on at most $m+1$ words, and for $m \geq 6$, one has $m+1 \leq F_m$. Hence Corollary 5.10 applies with $k = m+1$, yielding the lower bounds. □

B A Local Normalization Description of the Fold

This appendix records a local description of the canonical fold from Definition 2.1. It is included only to make the model fully explicit and reproducible.

Let $\mathcal{D}$ be the set of finitely supported digit configurations

$$a = (a_1, a_2, \dots), \qquad a_k \in \mathbb{N},$$

and define the Fibonacci value

$$\mathrm{Val}(a) := \sum_{k\geq 1} a_k F_{k+1}.$$

Embed $\omega = \omega_1 \cdots \omega_m \in \Omega_m$ as

$$\iota_m(\omega) := (\omega_1, \dots, \omega_m, 0, 0, \dots) \in \mathcal{D}.$$

Consider the following value-preserving local rewrites:

$$e_k + e_{k+1} \to e_{k+2}, \qquad 2e_1 \to e_2, \qquad 2e_2 \to e_1 + e_3, \qquad 2e_k \to e_{k-2} + e_{k+1} \quad (k \geq 3),$$

where e_k is the k-th unit vector in $\mathcal{D}$.

Proposition B.1 (Order-independent local normalization)**.** *Starting from $\iota_m(\omega)$, every maximal rewrite sequence terminates at the same normal form $\mathrm{NF}(\iota_m(\omega)) \in \{0,1\}^{\mathbb{N}}$ with no adjacent 1's. This normal form is the Zeckendorf expansion $Z(N(\omega))$. Consequently,*

$$\mathrm{Fold}_m(\omega) = \mathrm{pref}_m\big(\mathrm{NF}(\iota_m(\omega))\big)$$

is independent of rewrite order.

Proof. Each rewrite preserves Val because it encodes one of the Fibonacci identities

$$F_{k+1} + F_{k+2} = F_{k+3}, \qquad 2F_2 = F_3, \qquad 2F_3 = F_2 + F_4, \qquad 2F_{k+1} = F_{k-1} + F_{k+2} \quad (k \geq 3).$$

Termination follows from a lexicographic descent argument. Let

$$A(a) := \sum_{k\geq 1} a_k, \qquad B(a) := \sum_{k\geq 1} k\, a_k, \qquad C(a) := a_2.$$

The rules $e_k + e_{k+1} \to e_{k+2}$ and $2e_1 \to e_2$ decrease A; the rules $2e_k \to e_{k-2} + e_{k+1}$ for $k \geq 3$ preserve A and decrease B; and the rule $2e_2 \to e_1 + e_3$ preserves A and B but decreases C. Hence (A, B, C) decreases lexicographically at each step, so rewriting must terminate.

A normal form has no adjacent 1's and no digit ≥ 2, hence is a Zeckendorf representation. Since every rewrite preserves Val and Zeckendorf representation is unique [**?**], all terminating branches have the same normal form. The final formula for Fold_m is then just the definition of the canonical fold through prefix truncation. □

C Uniqueness of the Fibonacci Weighting

The preceding appendix shows that the canonical fold is implemented by Zeckendorf normalization. A natural question is whether the choice of Fibonacci weights F_{k+1} is itself forced. The following theorem shows that it is, within a natural axiom class.

Theorem C.1 (Contiguous numeration uniqueness)**.** *Let $w_2, w_3, \dots$ be positive integers. Define the valuation*

$$V_m^{(w)}(x_1 \cdots x_m) := \sum_{k=1}^{m} x_k\, w_{k+1}.$$

If for every $m \geq 1$ the map $V_m^{(w)}$ restricts to a bijection from X_m onto $\{0, 1, \dots, F_{m+2} - 1\}$, then $w_{k+1} = F_{k+1}$ for all $k \geq 1$.

Proof. For $m = 1$, $X_1 = \{0, 1\}$ and the bijection condition gives $\{0, w_2\} = \{0, 1\}$, so $w_2 = 1 = F_2$.

For $m = 2$, $X_2 = \{00, 01, 10\}$ and $\{0, w_2, w_3\} = \{0, 1, 2\}$; since $w_2 = 1$, one obtains $w_3 = 2 = F_3$.

Proceed by induction. Suppose $w_2 = F_2, \ldots, w_m = F_m$ for some $m \geq 3$. Decompose X_m by its last symbols:

$$X_m = \{u0 : u \in X_{m-1}\} \sqcup \{v01 : v \in X_{m-2}\}.$$

On the first part, $V_m^{(w)}(u0) = V_{m-1}^{(w)}(u)$. On the second, $V_m^{(w)}(v01) = V_{m-2}^{(w)}(v) + w_{m+1}$. By the inductive hypothesis,

$$V_{m-1}^{(w)}(X_{m-1}) = \{0, \ldots, F_{m+1} - 1\}, \quad V_{m-2}^{(w)}(X_{m-2}) = \{0, \ldots, F_m - 1\}.$$

Hence

$$V_m^{(w)}(X_m) = \{0, \ldots, F_{m+1} - 1\} \sqcup \{w_{m+1}, \ldots, w_{m+1} + F_m - 1\}.$$

For this to equal $\{0, \ldots, F_{m+2} - 1\} = \{0, \ldots, F_{m+1} + F_m - 1\}$, the second interval must start at F_{m+1}, giving $w_{m+1} = F_{m+1}$. □

Corollary C.2 (Uniqueness of the canonical residue fold)**.** *Let $G_m : \Omega_m \to X_m$ be a deterministic map satisfying:*

1. *$G_m(x) = x$ for all $x \in X_m$.*
2. *There exist positive integer weights w such that $V_m^{(w)}|_{X_m}$ is a bijection onto $\{0, \ldots, F_{m+2} - 1\}$.*
3. *For every $\omega \in \Omega_m$, $V_m^{(w)}(G_m(\omega)) \equiv \sum_{k=1}^{m} \omega_k w_{k+1} \pmod{F_{m+2}}$.*

Then $G_m = \mathrm{Fold}_m$.

Proof. Theorem C.1 forces $w_{k+1} = F_{k+1}$. Condition 2 then provides a unique legal representative in each residue class modulo F_{m+2}, and condition 3 selects that representative. This is exactly the residue characterization of Proposition 2.3. □